

Short-term wind speed prediction based on LMD and improved FA optimized combined kernel function LSSVM[☆]

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ABSTRACT

Accurate prediction of wind speed is of great significance to the operation and maintenance of wind farms, the optimal scheduling of turbines, and the safe and stable operation of power grids. A new prediction method for short-term wind speed based on local mean decomposition (LMD) and combined kernel function least squares support vector machine (LSSVM) is proposed. The short-term wind speed time series is decomposed into some components by the LMD algorithm. Based on LSSVM, radial basis function and the Polynomial function are used to generate the combined kernel function. The combined kernel function LSSVM combines the advantages of the radial basis function and the Polynomial function, which can achieve better prediction accuracy. The decomposed wind speed time series are predicted separately by the combined kernel function LSSVM model. At the same time, an improved firefly algorithm is proposed to optimize the parameters of the combined kernel function LSSVM. The final predictive value can be obtained by superimposing the predicted value of each combined kernel function LSSVM prediction model. The actual collected short-term wind speed data is chosen as the research object, the simulation experiments with four prediction horizons have been implemented. Compared with state-of-the-art prediction methods, through the comparison result curve between the prediction and actual wind speed, the box-plot results of predictive error distribution, the comparison results of the relative prediction error, the performance indicators, the Pearson's test, the DM test and the Taylor diagram results show that the proposed prediction method has higher prediction accuracy and is able to reflect the laws of wind speed correctly. Furthermore, the simulation results of four new datasets and adding noise to the input data of training set show that the proposed prediction method has strong robustness.

1. Introduction

Electricity belongs to a market that involves generation, transmission and distribution agents as well as consumers, which desire to maximize their profit and minimize expenses (Ribeiro et al., 2019). Because of the current energy shortage in the world, the contradiction between electricity supply and demand is prominent. Conventional energy sources such as oil and coal face the problem of insufficient reserves. As an important renewable natural energy, wind energy has the advantages of huge reserves, renewability and pollution-free, so it is recognized as an ideal renewable energy generation mode (Mishra et al., 2018; Tian et al., 2018a,b). The output power of wind power generation is closely related to wind speed of wind farm (Pradhan and Bhende, 2019). Wind speed shows strong randomness and rapid variation (Araghi et al., 2019). The accurate prediction of wind speed in wind farms is of great practical significance for power departments to adjust dispatching plans in time and to measure the capacity reliability of wind farms (Murthy and Rahi, 2017; Vargas et al., 2019). Generally,

the prediction of wind speed can be divided into long-term, medium-term and short-term (Wang et al., 2016). Long-term prediction is to predict future wind speed in the next few years. Medium-term prediction refers to the prediction of wind speed several days, weeks and months in advance. Short-term wind speed prediction usually requires 1 to 48 h in advance to predict wind speed. Medium-term and long-term wind speed prediction is mainly used for the evaluation of wind resources in order to facilitate the planning and construction of wind farms. Because of the randomness and uncertainty of wind speed, the prediction of short-term wind speed has more theoretical and practical value. Effective short-term wind speed prediction has important application value for wind power generation. The main determinant of wind power generation is local wind speed, so effective prediction of wind speed is the key factor to predict wind power (Hu et al., 2018).

It is well known that wind speed is affected by many parameters such as atmospheric pressure, temperature, humidity and etc. (Niau, 1995; Song et al., 2020). Therefore, if these environmental variables can be reasonably introduced into the prediction model, the prediction

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accuracy of short-term wind speed can be improved theoretically. However, most of the current wind speed prediction methods only use historical data of wind speed, rather than these environmental variables. The main reasons are as follows.

1. Many new sensors are needed to measure these environmental variables. The installation, testing and data transmission of sensors will increase the cost of wind speed acquisition system.
2. There is a certain coupling and correlation between these environmental variables. If these environmental variables do not use principal component analysis to extract principal components, they will be used as input of prediction model, which will make the information provided by these variables overlap to a certain extent, so that the prediction accuracy will be reduced.
3. These environmental variables are the input of the prediction model, which will increase the complexity of modeling, increase the training time of the model, and reduce the real-time performance of the prediction model.

At present, the commonly used wind speed prediction methods include physical prediction method, persistence method, time series method, Kalman filter, artificial neural network, fuzzy logic, deep learning machine, and combined prediction model, and etc. These research methods are discussed in detail below.

1. Prediction models based on physical methods mainly include numerical weather prediction (NWP) (Yang et al., 2018; Wang et al., 2019a), spatial correlation method (Ye et al., 2017), etc. The prediction model based on NWP can obtain wind speed prediction values for the next 1 to 3 days. Its prediction accuracy is relatively stable, but the calculation is huge. It often requires a huge computer to operate continuously for hours, so it is not suitable for short-term wind speed prediction. The spatial correlation method establishes a spatial correlation model based on the wind speed data of wind stations around wind farms to predict the wind speed in the coming hours. But, the spatial correlation method is greatly affected by the location of wind stations and the change of wind direction. Therefore, there are many limitations in the application of prediction models based on physical methods.

2. The persistence method considers that the predictive wind speed is equal to the sliding average value of the recent historical wind speed values (Santamaria-Bonfil et al., 2016). Usually, the wind speed value of the nearest sampling time is simply used as the wind speed prediction value at the next sampling time. The prediction error of the persistence method is large. And the predictive results are unstable.

3. Time series method uses a mass of historical data to build the model. Through model recognition, parameter estimation and model testing to determine a mathematical model that can describe the variation law of wind speed, and then deduces the prediction model to achieve the purpose of prediction. Time series method include auto regressive moving average (ARMA) model (Erdem and Shi, 2011), auto regressive integrated moving average (ARIMA) model (Cadenas et al., 2016), auto-regressive integrated moving average with exogenous variables (ARIMAX) model (Yan et al., 2017), and etc. However, there are some shortcomings, such as low prediction accuracy of low-order model and difficulty in parameter estimation of high-order model.

4. The most typical advantage of Kalman filter prediction is that it can correct the previous estimated value according to the latest measured value, which has the characteristics of dynamic weighting correction. Therefore, there are many short-term wind speed prediction models based on Kalman filter (Galanis et al., 2017; Cai et al., 2019; Zuluaga et al., 2015). In the calculation of observation vector and state vector in Kalman filter model, it is necessary to assume that the noise in the model is white noise with zero mean value. However, it is difficult to accurately determine the statistical characteristics of noise in the actual wind speed system. Because of the inaccuracy of the noise characteristics, the Kalman filter model cannot accurately describe the changing law of the wind speed system.

5. Artificial neural network is designed to imitate the structure and function of human brain. It is composed of a large number of simple processing elements connected in a large scale with a certain topological structure. It has the characteristics of parallel processing, distributed storage and fault tolerance, self-learning, self-organization, self-adaptation, and can be used for short-term wind speed prediction (Huseyin and Tansu, 2019). The typical used neural network for short-term wind speed include radial basis function neural network (Chang et al., 2017), fuzzy neural network (Ma et al., 2017), Elman neural network (Yu et al., 2018, 2017), extreme learning machine (ELM) (Sun et al., 2018; Luo et al., 2018; Tian et al., 2019), echo state network (Chitsazan et al., 2019), wavelet neural network (Santhosh et al., 2018; Wang et al., 2017), nonlinear neural network (Maruliya Begam and Deepa, 2019; Liu et al., 2018), and etc. However, neural network has inherent defects such as difficult to determine the network structure scientifically, slow learning speed, existence of local optimal value, memory instability and so on, which makes the prediction accuracy difficult to guarantee. On the other hand, the training of neural network needs a lot of sample data, which increase the computation time.

6. Fuzzy logic method uses fuzzy logic to form the professional knowledge of engineers and technicians into a fuzzy rule base (An et al., 2014; Khorramdel et al., 2018). Then a linear model is selected to approximate the wind speed with non-linear dynamic changes. However, for wind speed prediction, the effect of fuzzy logic method is often not good. This is mainly because the learning ability of fuzzy system is weak, and the identification of fuzzy system has not yet formed a perfect theory (Wang and Zhang, 2018). The process of fuzzification and the assignment of fuzzy rules are lack of systematic methods, which mainly rely on experience and trial implementation. Further research is needed to determine the structure of the fuzzy system in the forecasting system.

7. In recent years, many deep learning models are introduced to predict wind speed. These models include long short-term memory (LSTM) (Zhang et al., 2019a), convolutional neural networks (CNN) (Zhou et al., 2018; Mi et al., 2019), recursive neural network (RNN) (Li et al., 2019), and etc. However, deep learning models are very complex, and they need a lot of computing resources to model. Although the cost of calculation has been greatly reduced, the cost of calculation still needs to be considered. For the simple problem of small data sets, with the same computational overhead and time, deep learning models may not produce enough additional effects than the model described above.

8. Other scholars try to use the combined model to predict wind speed (Zhao et al., 2016; Kadhem et al., 2017; Wang and Li, 2019; Jiang et al., 2018; Zhou et al., 2019a; Moreno and Coelho, 2018; Wang et al., 2019b). The combined model is also named the hybrid or model. At present, the combined model has been widely used in many fields, such as finances (Zhou et al., 2019b), agribusiness (Jimenez-Donaire et al., 2020), health (Islam et al., 2019), industry (Wang et al., 2019d; Han et al., 2019), and etc. In these fields, the combined models often achieve better predictive performance than single models. Therefore, it is very important to study and apply the combined model. In the combined model, the advantages of different prediction models are combined, and good prediction results are achieved. However, it is difficult to determine the appropriate combined prediction model and the weight coefficients of each prediction model (Wang et al., 2019c; Bokde et al., 2019). At the same time, compared with the single prediction model, the advantages of the combined prediction model are difficult to prove and analyze in theory (Zhang et al., 2019b). Therefore, it is not guaranteed that the performance of the combined prediction model is better than that of the single prediction model.

In summary, most of these prediction methods are aimed at the original wind speed series, and the average relative prediction error usually reaches about 10% to 20%. Obviously, the limited short-term wind speed data itself cannot completely reproduce all the characteristics of the wind system, but these observed wind speed data can be regarded as a kind of signal, which can be processed to obtain a lot of

useful information. Because wind speed is influenced by temperature, air pressure, topography and other factors, it has strong randomness and contains very rich characteristic information. By preprocessing the original wind speed series, the information of different characteristics is decomposed. This will reduce the difficulty of modeling and hopefully improve the prediction accuracy. Local mean decomposition (LMD) proposed in 2005 provide a good solution to the problem of information extraction in the original signal (Smith, 2005). LMD is highly adaptive. It can decompose any complex non-stationary signal to multiple and physically meaningful production function (PF) components (Jia et al., 2019). Then the instantaneous frequency and amplitude of the obtained PF component are combined to obtain the complete information distribution of the original signal (Yu and Lv, 2017). If the factors affecting wind speed and their influencing rules remain basically unchanged, the PF components obtained by LMD decomposition can better describe the time evolution law of one or more of the influencing wind speed changes. Therefore, LMD can lay a good foundation for extracting the relevant information of the internal variation law of wind speed.

In 2000, based on support vector machine (SVM), least squares support vector machine (LSSVM) was proposed. LSSVM transforms the quadratic programming problem in SVM into solving linear equations, which reduces the complexity of solving. Therefore, LSSVM is very suitable for short-term wind speed prediction and has achieved some research results (Wang et al., 2018; Xiang et al., 2019). One of the greatest advantages of LSSVM algorithm is that it needs fewer samples and ensures higher prediction accuracy (Chen et al., 2019). In practical application, the samples of short-term wind speed are sometimes difficult to obtain or there are many unqualified samples, so the sample size is very small. Therefore, the practicability of LSSVM has more advantages than neural network, deep learning model, and etc. The combination of the decomposed PF component and the small sample advantage of LSSVM model can fully explore the physical mechanism and laws contained in the short-term wind speed data itself, which is also in line with the mechanism of using data self-driving to obtain the influencing factors, thus laying a good foundation for improving the prediction accuracy of short-term wind speed. However, the prediction accuracy of LSSVM is affected by the kernel function and the parameters of the kernel function (Liu et al., 2017). At present, there is no definite theory and method to support how to determine the kernel function and the parameters of the kernel function. In order to improve the prediction accuracy of LSSVM and solve the problem of selecting the kernel function and the parameters of the kernel function in LSSVM algorithm, this paper uses global kernel function and local kernel function to construct the combined kernel function. The global kernel function is used to fit the distant correlation in the sample, and the local kernel function is used to approximate the correlation in the adjacent data domain, so as to improve the generalization ability of the LSSVM model. In order to determine the optimal parameters of the combined kernel function LSSVM, an improved firefly algorithm (FA) is proposed to optimize the parameters. This improved FA improves the weight update strategy of standard FA to accelerate the convergence speed of the population and reduce the impact of randomness on the performance of the algorithm.

Based on the above description, this paper first uses LMD to process the short-term wind speed, and obtains the PF components and residual component of the original short-term wind speed series. Then, each PF component and residual component is trained and predicted by combined kernel function LSSVM model. The parameters of each combined kernel function LSSVM model are optimized by improved FA. Finally, the prediction results of different PF components and residual component are superimposed to generate final short-term wind speed prediction values.

The main innovations and research ideas of this paper are given.

1. LMD is introduced to obtain the PF components and residual component of original short-term wind speed time series.

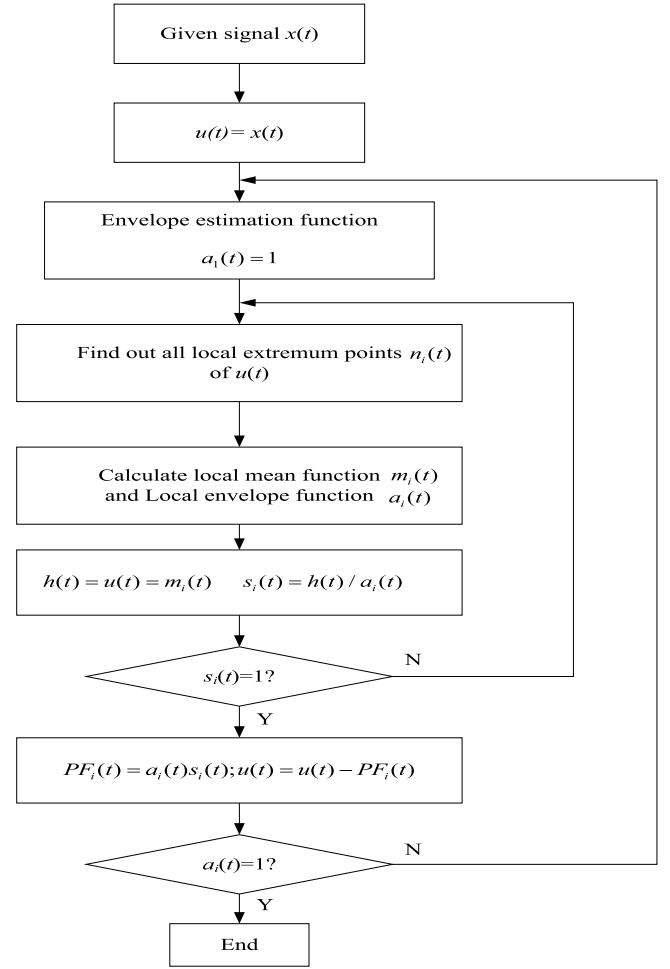


Fig. 1. The specific decomposition steps of LMD algorithm.

2. The combined kernel function LSSVM model is proposed to improve the prediction ability. The combined kernel function LSSVM is chosen as the prediction model of each PF and residual component.

3. An improved FA is proposed to optimize the parameters of combined kernel function LSSVM model.

Compared with traditional prediction models such as neural network and time series, and modern computational intelligence tools such as LSTM and CNN, the advantages of the proposed prediction method are summarized as follows.

1. The algorithm of the proposed prediction method is LSSVM. Therefore, the prediction method needs fewer samples and has more advantages in practice. This is the main advantage of the proposed prediction method.
2. Compared with the deep learning model and the neural network, the training time of the prediction method in this paper is less.
3. The combined kernel function LSSVM model has low dependence on single kernel functions and their parameters, and has high modeling accuracy. Combined with improved FA, it is more reasonable to determine the parameters of the prediction model.
4. LMD can effectively explain the fluctuation, periodicity and trend of short-term wind speed.

The content of the paper is arranged as follows: Section 1 reviews the current status of short-term wind speed prediction; Section 2 outlines the theory of the LMD, combined kernel function LSSVM, and improved FA; Section 3 introduces the implementation steps of the proposed prediction method for short-term wind speed prediction; The

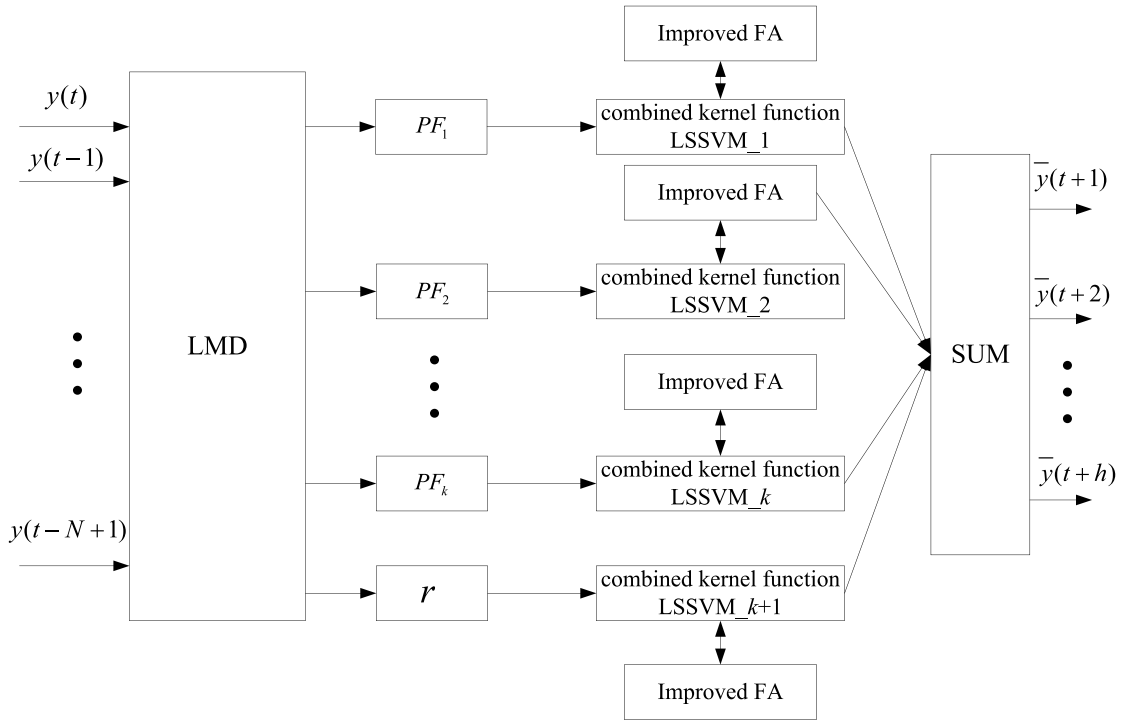


Fig. 2. The structure of the proposed prediction method for short-term wind speed.

experimental results are provided in Section 4, and the validity of the proposed prediction method in this paper is shown; The discussion based on the comparison results is presented in Section 5; Finally, Section 6 concludes the paper.

2. Basic theories

In this section, the required theories that are applied in the developed prediction framework are introduced as follows.

2.1. LMD

LMD is a nonlinear signal analysis method proposed in 2005. Its essence is to decompose the signal adaptively into the sum of a series of PF components with physical significance, thus reflecting the intrinsic nature of the signal and obtaining the time–frequency distribution of the original signal. Many signals in practical engineering applications are nonlinear and non-stationary. Using the ensemble decomposition method can reduce the non-stationary of the signal and extract the information of different frequency. On the basis of this, the ensemble decomposition method is combined with some prediction methods or prediction models can improve the prediction effect and achieve better performance index than single model. In these ensemble decomposition methods, LMD, empirical mode decomposition (EMD), ensemble empirical mode decomposition (EEMD), complementary empirical mode decomposition (CEEMD) are typical representatives. Compared with EMD, EEMD and CEEMD, LMD has some unique advantages. EMD has the phenomenon of end point overlapping. EEMD and CEMD add the white noise in the original signal, which will increase the calculation time. LMD can decompose the signal adaptively according to its own characteristics, so that each component after decomposition has a certain physical significance, and can reflect the inherent nature of the signal, which provides a good foundation for other post-processing of the signal. At the same time, LMD has some advantages in restraining the end effect, reducing the calculation time and preserving the integrity of signal information.

For any original signal $x(t)$, the specific decomposition steps of LMD algorithm are as shown in Fig. 1.

It can be seen from Fig. 1 that LMD algorithm is a process of multiple cycles. LMD algorithm can extract pure frequency modulation signal and envelope signal from the original signal. A PF component can be obtained by multiplying these two kinds of signals. Then, all PF components can be obtained by gradually cycling, and the instantaneous frequency and amplitude can be finally obtained. The instantaneous frequency can be obtained from the pure frequency modulation signal. The instantaneous amplitude represents the amplitude modulation information of PF component, and the instantaneous frequency represents the frequency modulation information of PF component. Finally, the complete time–frequency distribution of the original signal is obtained. In Fig. 1, $x(t)$ is a given signal, $h(t)$ and $u(t)$ are intermediate variables, $a_i(t)$ is envelope function, $PF_i(t)$ is production function component, $s_i(t)$ is pure frequency component, $n_i(t)$ is local extremum point, $m_i(t)$ is local mean function. By LMD algorithm, the original signal $x(t)$ is decomposed into k PF components and a new residual signal $r_k(t)$.

$$x(t) = \sum_{n=1}^k PF_n(t) + r_k(t) \quad (1)$$

where k is the number of decomposition.

2.2. Combined kernel function LSSVM

Through a nonlinear mapping function, the sample space of LSSVM is mapped into a high-dimensional or even infinite dimensional feature space (Chen and Zhou, 2018). For a sample dataset $\{x_i, y_i\}, i = 1, 2, \dots, N$, there is

$$y = w\varphi(x) + b \quad (2)$$

where, N is the total number of samples; $y, y \in \mathbf{R}^d$ is output of nonlinear systems; $x, x \in \mathbf{R}^m$ is input of nonlinear systems; m is the dimension of sample space; $\varphi()$ is a mapping function of input space with m -dimensional to output space with high d -dimensional ($d \gg m$), $w, w \in \mathbf{R}^d$ is the weight coefficient vector, b is the constant bias. Optimal w and b can be obtained by minimizing the next objective function.

$$\min_{w,b,e} J(w, e) = \frac{1}{2} w^T w + \frac{1}{2} \gamma \sum_{i=1}^N e_i^2 \quad (3)$$

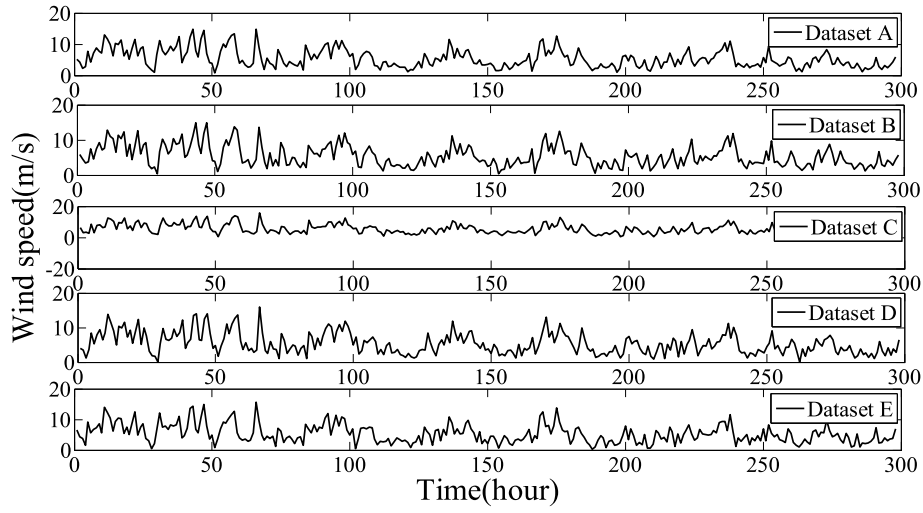


Fig. 3. Five short-term wind speed datasets.

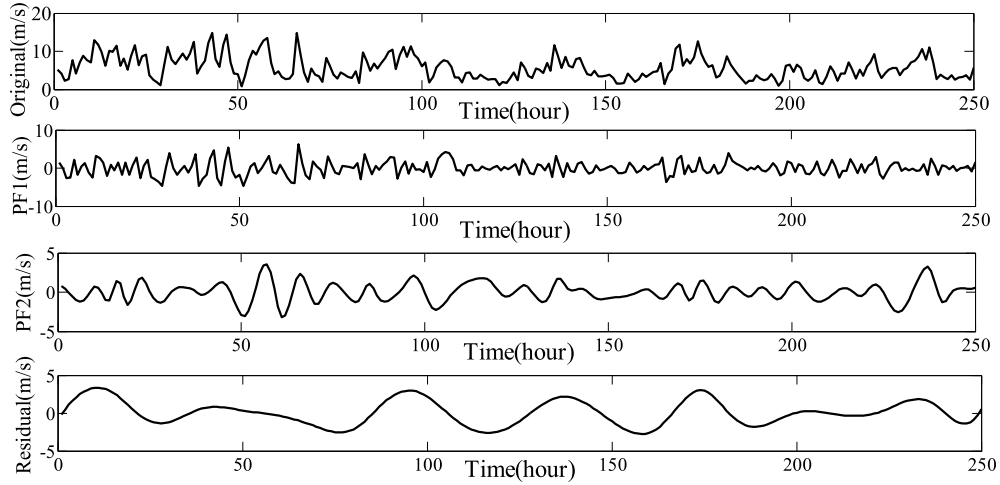


Fig. 4. The original short-term wind speed, PF components, and residual component.

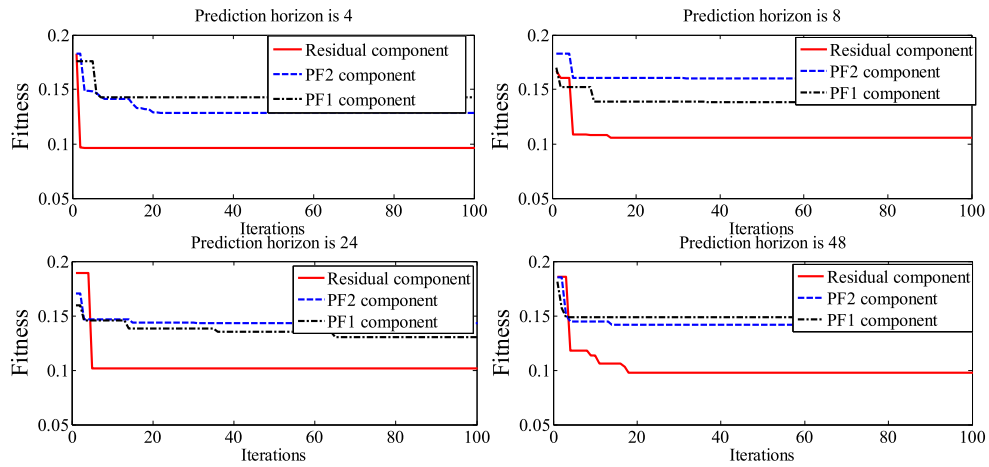


Fig. 5. The fitness curve of 3 components with four prediction horizons.

where, $J(\mathbf{w}, e)$ represents structural risk. $e_t, t = 1, 2, \dots, N$ is permissible error, it means the predictive error between actual and predictive output. γ is regularization parameter that controls the degree of penalty for the error.

Lagrange function is introduced to solve the constrained optimization problems mentioned above,

$$L(\mathbf{w}, b, e, a) = J(\mathbf{w}, e) - \sum_{t=1}^N a_t (\mathbf{w}^T \varphi(\mathbf{x}_t) + b + e_t - y_t) \quad (4)$$

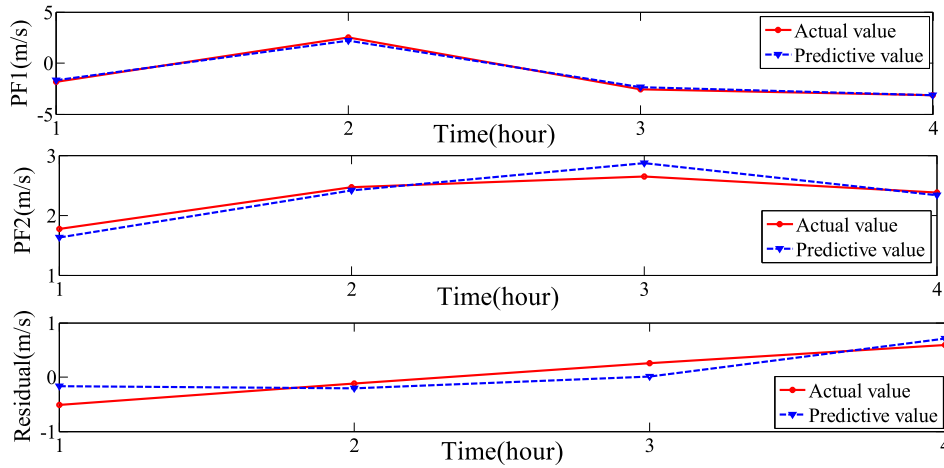


Fig. 6. The comparison curves between the actual values and the predictive values of each component when prediction horizon is 4.

where, a_i is a Lagrange multiplier. The Lagrange function is employed to make the above optimization problem convert to solve the linear equations. According to Mercer conditions, the presence of mapping function $\varphi()$ and kernel function $K()$ satisfy the following equation,

$$K(\mathbf{x}_i, \mathbf{x}_j) = \varphi(\mathbf{x}_i)^T \varphi(\mathbf{x}_j) \quad (5)$$

Nonlinear mapping ability of LSSVM prediction model is determined by the kernel function. Kernel function is used for mapping samples of LSSVM prediction model from an input space to a feature space. Therefore, different kernel function has different learning ability and generalization ability. There are many kernel functions, which can be summarized into two categories: partial kernel and global kernel functions. In the LSSVM model using global kernel function, the farther from the test point, the larger the kernel value is. On the contrary, in the LSSVM with partial kernel function, the closer the test point is, the smaller the kernel function value is. Polynomial kernel function is a typical representative of global kernel functions, and radial basis function (RBF) is a typical representative of partial kernel functions. Polynomial kernel function is indicated in Eq. (6). RBF kernel function is indicated in Eq. (7).

$$K(\mathbf{x}_i, \mathbf{x}_j) = [\mathbf{x}_i \cdot \mathbf{x}_j + 1]^q \quad (6)$$

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}\right) \quad (7)$$

where, q is the order of Polynomial function, σ^2 is the width of RBF kernel function. The prediction value of LSSVM model is

$$\mathbf{y} = \sum_{i=1}^N \alpha_i K(\mathbf{x}_i, \mathbf{x}_j) + b \quad (8)$$

The introduction of kernel function avoids the complex calculation in high dimensional feature space, but different kernel functions achieve the different performance. The choice of kernel function becomes one of the core problems of LSSVM. The partial kernel function has a great influence on the data points in the small area near the test point, so it has strong learning ability, but the generalization ability is weak. The characteristics of global kernel function are just the opposite. That is meant the partial kernel function has strong learning ability and weak generalization performance, while global kernel function has strong generalization performance and weak learning ability. Therefore, we can consider the advantages of combining these two kinds of kernel functions, and construct the mixed-kernel function.

In the literature (Smits and Jordaan, 2002), the author points out that the Polynomial kernel function is a global kernel function with global characteristics and good generalization ability. The sample point

is far away from the test point will also affect the output of the function. As a kind of partial kernel function, the parameter σ^2 of RBF determine the width of the kernel function. Only the sample point close to the test point has an effect on the output of the function, so RBF function has partial characteristics and good interpolation ability. Therefore, in order to make the learning ability and generalization ability of LSSVM better, these two kernel functions can be combined to construct a combined kernel function. The advantages of two kernel functions are combined. Better regression prediction performance can be obtained. This paper uses the following method to construct a new combined kernel function.

$$K_{hybrid} = aK_{rbf} + (1 - a)K_{poly}, \quad a \in [0, 1] \quad (9)$$

where, K_{poly} is a polynomial kernel function expressed as Eq. (6), K_{rbf} is a RBF kernel function expressed as Eq. (7), a is weight coefficient. Because of K_{poly} and K_{rbf} satisfy the Mercer condition, their linear combinations K_{hybrid} also satisfy the Mercer condition. The weight coefficient a determines the ratio of a single kernel to a mixed-kernel. When a is greater than 0.5, the Polynomial kernel function is dominant. When a is less than 0.5, the RBF function is dominant. When a is equal to 0.5, two kernel functions are equally important. The flexible combination of RBF kernel function and Polynomial kernel function can be realized by adjusting a when constructing the combined kernel function. Therefore, the combined kernel function constructed in this paper can realize the advantages of the RBF kernel function in the fitting degree, and can also adopt the advantage of the generalization ability of the Polynomial kernel function. The combined kernel function combines the global properties of the Polynomial kernel function and the partial properties of the RBF kernel function, and has better generalization ability. But in the combined kernel function, each parameter has an impact on the ability of LSSVM. These parameters are interrelated and affected each other. These parameters include a , q , σ^2 and γ . In order to determine the optimal parameters of combined kernel function LSSVM prediction model, this paper proposes an improve FA in the next Section to solve this problem.

2.3. Improved FA

Evolutionary algorithms have shown to be effective in solving optimization problems in many engineering fields (Neto et al., 2018). FA is proposed by simulating the population behavior of natural fireflies (Yang, 2009). In the optimization process of the algorithm, firefly individuals are randomly distributed to the solution space of the objective function, and the stronger luminance the firefly itself is, the better the spatial position of the objective function is. Individuals of stronger luminance fireflies have stronger attractiveness. In the process

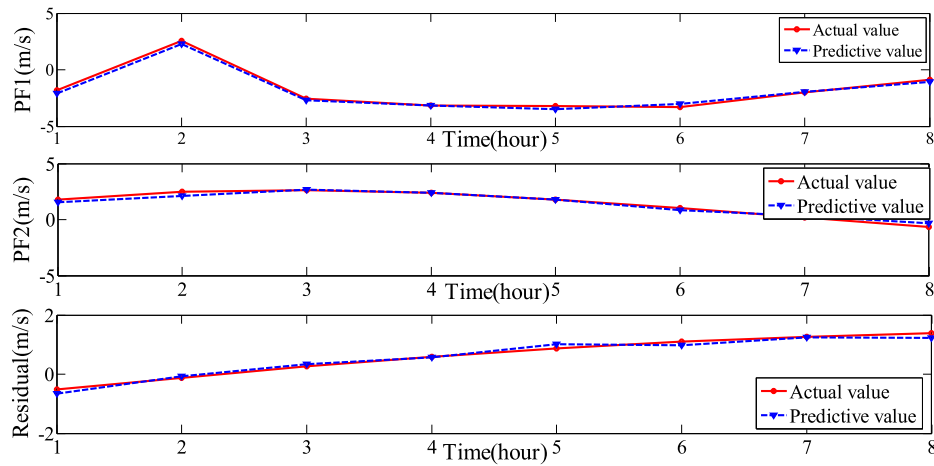


Fig. 7. The comparison curves between the actual values and the predictive values of each component when prediction horizon is 8.

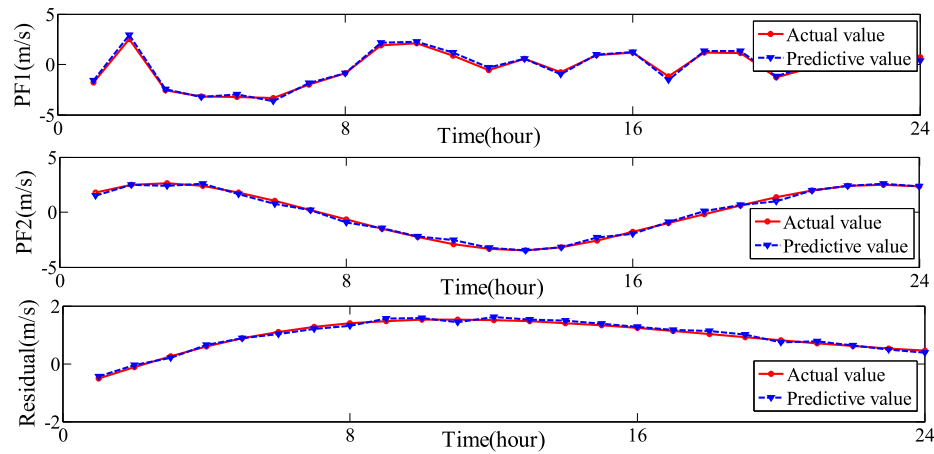


Fig. 8. The comparison curves between the actual values and the predictive values of each component when prediction horizon is 24.

Table 1

The average errors obtained from the global optimum and the standard deviation of four optimization algorithms.

Algorithm	F_1	F_2	F_3
Improved FA	$1.157\text{e}-03 \pm 1.12\text{e}-02$	$4.223\text{e}+01 \pm 4.264\text{e}+01$	$2.028\text{e}+03 \pm 4.257\text{e}+02$
Standard FA	$3.024\text{e}-02 \pm 1.787\text{e}-02$	$6.115\text{e}+02 \pm 3.191\text{e}+02$	$4.990\text{e}+03 \pm 3.367\text{e}+03$
ABC	$2.367\text{e}-03 \pm 1.441\text{e}-02$	$9.477\text{e}+01 \pm 6.115\text{e}+01$	$4.601\text{e}+03 \pm 3.007\text{e}+03$
PSO	$2.055\text{e}-03 \pm 1.118\text{e}-02$	$1.708\text{e}+02 \pm 1.005\text{e}+02$	$0.326\text{e}+04 \pm 1.118\text{e}+03$

Table 2

Statistical values of five datasets.

Dataset	Maximum value	Minimum value	Mean value	Standard deviation	Skewness	Kurtosis	Median	Sample size
<i>Dataset A</i>								
All samples	14.7900	0.7300	5.4031	2.9339	0.8934	3.3151	4.7000	298
Training set	14.7900	0.7300	5.6824	3.0328	0.7677	3.0153	5.1950	250
Testing set	9.3950	1.1450	3.9485	1.7581	0.8920	3.8250	3.7175	48
<i>Dataset B</i>								
All samples	14.9505	0.3863	5.4139	3.0154	0.8066	3.1213	4.7410	292
Testing set	9.7609	1.2531	3.9671	1.9150	1.0612	3.7720	3.5025	48
<i>Dataset C</i>								
All samples	15.7914	0.1077	5.3961	2.9611	0.8472	3.3638	4.7354	292
Testing set	9.6680	0.1077	4.0921	2.0394	0.7516	3.6255	3.5563	48
<i>Dataset D</i>								
All samples	16.0299	0.0300	5.3839	3.0673	0.8160	3.2751	4.8684	292
Testing set	9.0567	0.0300	3.9253	1.9316	0.4562	2.8210	3.7707	48
<i>Dataset E</i>								
All samples	15.7378	0.2903	5.3384	3.0327	0.8134	3.4658	4.9363	292
Testing set	9.7780	0.9292	4.0233	2.0108	0.6229	2.9957	3.7296	48

of inter-attraction, the stronger luminance of fireflies decreases with the increase of space distance and the absorption (Sanchez et al., 2017). In each iteration, the location of the firefly is constantly updated. In each iteration, the solution of the problem is further optimized. FA has the advantages of simple principle, few parameters, easy implementation, strong global optimization ability and convergence ability (Kumar et al., 2017; Durbhaka et al., 2019). From a mathematical point of view, the optimization process of FA is described as follows.

First, the relationship between firefly i 's absolute luminance I_i and objective function value is built. The luminance with a distance of r from the individual is generally expressed as

$$I = I_0 e^{-\theta r_{ij}} \quad (10)$$

I_0 is the luminance of the firefly itself.

$$I_0 = \frac{1}{1 + g(\eta)} \quad (11)$$

Suppose firefly i has a bigger absolute luminance than firefly j , j is attracted by firefly i and thus moves towards i , then to j , i 's attractiveness β_{ij} is

$$\beta_{ij} = \beta_0 e^{-\theta r_{ij}^2} \quad (12)$$

where, β_0 is the maximum attractiveness, θ is the absorption coefficient of light. β_0 generally take 1. $\theta \in [0.01, 100]$. r_{ij} is the cartesian distance from i to j . That is

$$r_{ij} = \|\eta_i - \eta_j\| = \sqrt{\sum_{k=1}^d (\eta_{ik} - \eta_{jk})^2} \quad (13)$$

where d is the dimension of variable.

Because j moves towards i under the attraction of i , then the update formula of j 's position is

$$\eta_j(t+1) = \eta_j(t) + \beta_{ij}(\eta_i(t) - \eta_j(t)) + \alpha \varepsilon_j \quad (14)$$

where t is the number of iterations; η_i and η_j are the spatial positions of i and j . β_{ij} is i 's relative attractiveness to j , α is a constant generally within $[0, 1]$, ε_j is the vector of random number obtained through uniform distribution.

For the standard FA, in the later stage of iteration, as the distance between fireflies decreases, the relative attractiveness between fireflies increases gradually, so FA's local search ability weakens and even has repeated oscillations near the extreme point, and thus may need multiple iterations to meet the required precision (Chen and Han, 2018). In this case, the algorithm may fail to meet the precision requirement with the limited number of iterations. To improve FA's local and global search abilities, it is considered to introduce the inertia weight into the position update formula. The following is the formula with the inertia weight

$$\eta_j(t+1) = \omega(t)\eta_j(t) + \beta_{ij}(\eta_i(t) - \eta_j(t)) + \alpha[rand - 0.5] \quad (15)$$

A weight ω too big or too small may affect the speed and precision of parameter estimation. The adaptive variable weight adopts a big ω in the earlier stage to enhance the global search ability and a small ω in the later stage to focus on local search and enhance the algorithm's local search ability, thus avoiding the repeated oscillations near the extreme point and increasing the precision of solution. For this reason, the paper uses an inertia weight nonlinear decreasing strategy allowing $\omega(t)$ decreasing nonlinearly as the number of iterations t increases. That is

$$\omega(t) = [1 - (\frac{t}{t_{\max}})^{\alpha_1}]^{\alpha_2} (\omega_{\max} - \omega_{\min}) + \omega_{\min} \quad (16)$$

where, t is the current number of iterations, $t_{\max}$ is the maximum number of iterations, $\omega_{\max}$ is the maximum value of inertia weight, $\omega_{\min}$ is the minimum value of inertia weight, and α_1 and α_2 are constants set initially. $\alpha_1 \in [0, 1]$, $\alpha_2 \in [1, +\infty]$.

To accelerate the convergence in population and reduce the effect of randomness, the paper proposes a update mode of α , which allows α decreases with the number of iterations. The following is the update equation

$$\alpha = (\alpha_{\max} - \alpha_{\min})e^{-d_1(\frac{t}{t_{\max}})^{d_2}} + \alpha_{\min} \quad (17)$$

where t is the number of iterations. $d_1 \geq 0$ and $d_2 \geq 0$.

In order to test the performance of the improved FA, the following three functions include shifted Sphere function (F_1), shifted Rosenbrock's function (F_2), and shifted Schwefel's problem (F_3) are selected for testing (Pierezan and Dos Santos Coelho, 2018). The definition and parameters of these three Benchmark functions are as the following.

$$F_1(x) = \sum_{i=1}^{30} z_i^2 - 450, z = x - o, x = [x_1, x_2, \dots, x_{30}], x \in [-100, 100]^{30} \quad (18)$$

$$F_2(x) = \sum_{i=1}^{49} (100(z_i^2 - z_{i+1})^2 + (z_i - 1)^2) + 390, z = x - o + 1, x = [x_1, x_2, \dots, x_{50}], x \in [-100, 100]^{50} \quad (19)$$

$$F_3(x) = \sum_{i=1}^{30} (\sum_{j=1}^i z_j)^2 - 450, z = x - o, x = [x_1, x_2, \dots, x_{30}], x \in [-100, 100]^{30} \quad (20)$$

These functions have been tested and compared with the results obtained by artificial bee colony (ABC), the particle swarm optimization (PSO), the standard FA. The ABC's parameters trials and food sources have been both setup as 100 (with one scout and 50% of employed bees). The PSO's parameters swarm size, cognitive constant, social constant and maximum speed have been setup as 100, 2, 2 and 20% of the search space. The inertia weight has been setup from 0.9 to 0.4 with linear decreasing. The standard FA's parameters population size, randomness and absorption coefficient have been setup respectively as 100, 0.2 and 0.1. The improved FA's parameters population size as 100, $\omega_{\max}$ is 1, $\omega_{\min}$ is 0.01, α_1 is 0.8, α_2 is 5, $\alpha_{\max}$ is 1, and $\alpha_{\min}$ is 0.01. The algorithms have been tested 30 times for each function with stopping criterion (maximum number of iterations) equals to 10000. The average errors obtained from the global optimum and the standard deviation are shown in Table 1. The software is Matlab R2010b. All the programs are coded in m-file. It can be seen from Table 1, the improved FA has reached lower errors for three Benchmark functions.

3. The implementation steps of the proposed prediction method for short-term wind speed

The structure of the proposed prediction method is shown in Fig. 2. In Fig. 2, LMD algorithm is decomposition unit, PF_i is the i th PF component obtained by LMD decomposition, r is residual component, SUM is predictive value superposition unit. Each individual combined kernel function LSSVM prediction model corresponds to each PF component and residual component.

It can be seen from Fig. 2, the inputs of the proposed prediction method is short-term wind speed $y(t), y(t-1), \dots, y(t-N+1)$, t is the current sampling time, N is the length of short-term wind speed. Using the proposed prediction method, the output is the future h short-term wind speed data.

Combined with Fig. 2, the steps of the prediction method for short-term wind speed proposed in this paper can be described as follows.

3.1. Modeling process

Step 1 Given a short-term wind speed time series $y(1), y(2), \dots, y(N)$, N is the length of the time series. The original short-term wind speed

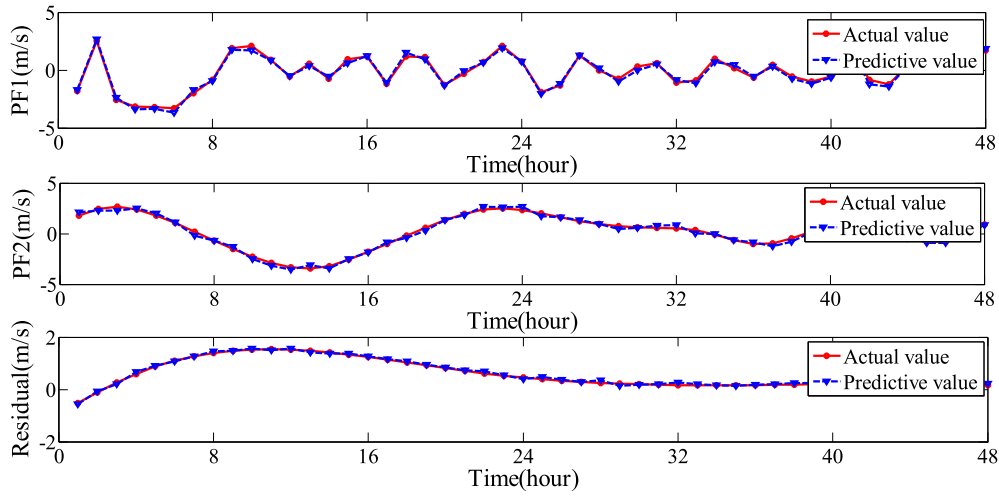


Fig. 9. The comparison curves between the actual values and the predictive values of each component when prediction horizon is 48.

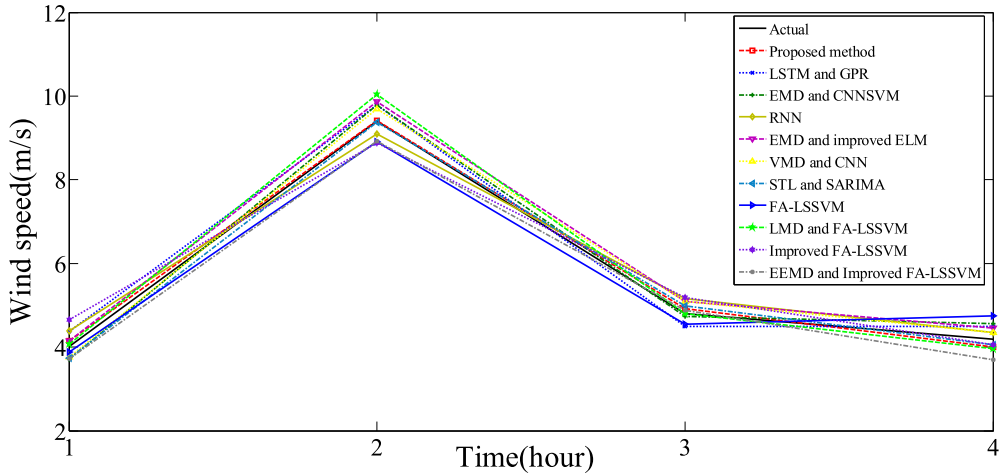


Fig. 10. The predictive comparison results of these prediction methods (prediction horizon is 4).

time series is decomposed into k PF components with different scales and 1 residual component by LMD.

Step 2 For each PF component and residual component, the proposed combined kernel function LSSVM is adopted as the prediction model. Assume the PF component and residual component after decomposition is expressed as $x(1), x(2), \dots, x(N)$, m is the window in the time series, the input vector of LSSVM model is generated as the Eq. (21). The output of LSSVM model is expressed as Eq. (22). Assume the prediction horizon is h . The input vector $\mathbf{X}$ and output vector $\mathbf{Y}$ are used to train the each combined kernel function LSSVM. Each row of matrix $\mathbf{X}$ in Eq. (21) is the input of the prediction model, and the corresponding row of matrix $\mathbf{Y}$ in Eq. (22) is the output of the prediction model. For example, $x(1), x(2), \dots, x(m)$ is input, then the output is $x(m+1), x(m+2), \dots, x(m+h)$.

$$\mathbf{X} = \begin{bmatrix} x(1) & x(2) & \cdots & x(m) \\ \vdots & \vdots & \ddots & \vdots \\ x(N-m) & x(N-m-1) & \cdots & x(N-1) \end{bmatrix} \quad (21)$$

$$\mathbf{Y} = \begin{bmatrix} x(m+1) & x(m+2) & \cdots & x(m+h) \\ \vdots & \vdots & \ddots & \vdots \\ x(N-m+h) & x(N-m+h+1) & \cdots & x(N-m+h-1) \end{bmatrix}^T \quad (22)$$

Step 3 For each prediction model, improved FA is used to optimize the parameters of combined kernel function LSSVM. The parameters

to be optimized are: the order of Polynomial function (q), the weight coefficient (a), the regularization parameter (γ), RBF kernel function width (σ^2), and the window (m). The range of parameters are as $a \in [0, 1]$, $q \in [1, 5]$, $\sigma^2 \in [0.01, 1000]$, $\gamma \in [0.01, 1000]$, and $m \in [1, 30]$. The parameters of improved FA include population size p , maximum number of iterations $t_{\max}$, $\omega_{\max}$, $\omega_{\min}$, α_1 , α_2 , $\alpha_{\max}$, and $\alpha_{\min}$ are initialized. Let the current iteration number i is 1.

Step 3.1 Initialize the positions of fireflies randomly. The maximum attractiveness β_0 , the absorption coefficient θ are determined.

Step 3.2 Calculate objective function value and firefly's luminance I . The fitness function is chosen as the following Eq. (23).

$$f_i = \frac{1}{h} \sqrt{\sum_{k=1}^h (x(k) - \hat{y}(k))^2} \quad (23)$$

where $x(k)$ is actual value of each component, $\hat{y}(k)$ is predictive value of each component, h is prediction horizon.

Step 3.3 According to Eq. (16), adaptive variable weight $\omega(t)$ is calculated. According to Eq. (17), α is determined. Compare each firefly with all the others in terms of initial luminance and make position update when meeting a firefly with the stronger luminance. The position update is performed out according to Eq. (15).

Step 3.4 If the maximum number of iterations is satisfied, go to **Step 4**; otherwise, $i = i + 1$, go to **Step 3.2**.

Step 4 Output the optimal parameters. Repeat the train and optimization process, until all combined kernel function LSSVM models are determined.

Table 3

The detailed parameters of these prediction methods.

Prediction method	Symbol	Meaning	Value	Reason
LSTM and GPR	n_i	Number of input layer nodes	250	Number of inputs
	n_h	Number of hidden layer nodes	8	Common value [2, 4, 8, 10, ...]
	n_o	Number of output layer nodes	4, 8, 24, 48	Prediction horizons
	η	Fixed learning rate	0.01	Common value [0.001, 0.01, 0.05, 0.1, ...]
	T	Size of batch	32	Common value [8, 16, 32, 50, 100, ...]
	E_p	Epochs of training	2000	Converged
	p_1	Kernel function	2	Obtained by GA in [-5, 5]
	p_2	Parameter in Gaussian Function	1	Obtained by GA in [-5, 5]
EMD and CNN SVM	k_f	Kernel function	Gaussian Function $k_{ij} = p_1 \exp(-\frac{(x_i - x_j)^2}{2p_2})$	A competitive kernel function
	N_{imf}	The number of components	5	Obtained by EMD
	f	Activation function	Rectified linear unit (ReLU)	A competitive kernel function
	p_1	The ratio of the dropout	0.7	Common value
	p_2	The proportion of the least absolute shrinkage and selection operator (LASSO) regularization	0.01	Common value
	C_L	The channels in the convolutional layers	4, 8, 16	Common value
	C	Regularized constant of SVM	Prediction horizon is 4 (IMF1-3.324; IMF2-7.205; IMF3-6.207; IMF4-11.365; IMF5: 0.365) Prediction horizon is 8 (IMF1-17.357; IMF2-9.990; IMF3-25.304; IMF4-7.027; IMF5: 6.217) Prediction horizon is 24 (IMF1-7.707; IMF2-13.022; IMF3-1.357; IMF4-9.995; IMF5: 0.802) Prediction horizon is 48 (IMF1-14.204; IMF2-3.332; IMF3-8.227; IMF4-4.327; IMF5: 6.340)	Obtained by grid-search method
	ε	The insensitive loss factor of SVM	Prediction horizon is 4 (IMF1-1.364; IMF2-0.997; IMF3-0.654; IMF4-2.382; IMF5: 4.364) Prediction horizon is 8 (IMF1-16.328; IMF2-3.654; IMF3-3.775; IMF4-6.327; IMF5: 5.004) Prediction horizon is 24 (IMF1-10.320; IMF2-6.328; IMF3-9.334; IMF4-1.220; IMF5: 16.302) Prediction horizon is 48 (IMF1-10.220; IMF2-4.005; IMF3-14.227; IMF4-5.007; IMF5: 3.447)	Obtained by grid-search method
RNN	n_i	Number of input layer nodes	250	Number of inputs
	n_L	LSTM layer	200	Common value
	n_{F1}	Fully connected layer 1	50	Common value
	p_1	The ratio of the dropout	0.7	Common value
	n_{F2}	Fully connection layer	12 regression layer is 1	Common value
	n_o	Number of output layer nodes	4, 8, 24, 48	Prediction horizons
EMD and improved ELM	N_{imf}	The number of components	5	Obtained by EMD
	m	Embedding dimension	20	Trial and error method
	L	Number of hidden layer	100	Trial and error method
	μ	The weight coefficient	0.9	Common value
	λ	Parameter of activation function	1	Common value
	ε	The threshold value	0.2	Common value
	k_f	Activation function	Sigmoid function $k(x) = \frac{1}{1+e^{-\lambda x}}$	A competitive kernel function
VMD and CNN	N_{imf}	The number of components	4	Obtained by VMD
	n_i	Number of input layer nodes	250	Number of inputs
	C_1	Convolutional layer 1	48	Common value
	C_2	Convolutional layer 2	32	Common value
	C_3	Convolutional layer 3	16	Common value
	C_4	Convolutional layer 4	16	Common value
	n_o	Number of output layer nodes	4, 8, 24, 48	Prediction horizons
	f	Activation function	Rectified linear unit (ReLU)	A competitive kernel function

(continued on next page)

Table 3 (continued).

Prediction method	Symbol	Meaning	Value	Reason
STL and SARIMA	p	Autoregressive coefficient	Prediction horizon 4: 2, 2, 3	Obtained by Akaike information criterion (AIC)
			Prediction horizon 8: 2, 2, 2	
			Prediction horizon 24: 2, 2, 3	
			Prediction horizon 48: 2, 2, 2	
	q	Moving average coefficient	Prediction horizon 4: 1, 1, 1	Obtained by Akaike information criterion (AIC)
			Prediction horizon 8: 1, 1, 1	
			Prediction horizon 24: 1, 1, 2	
			Prediction horizon 48: 1, 2, 1	
	d	Difference coefficient	Prediction horizon 4: 3, 1, 2	Obtained by Akaike information criterion (AIC)
			Prediction horizon 8: 2, 2, 1	
			Prediction horizon 24: 3, 3, 1	
			Prediction horizon 48: 4, 1, 1	
	P	Seasonal autoregressive coefficient	Prediction horizon 4: 1, 1, 1	Obtained by Akaike information criterion (AIC)
			Prediction horizon 8: 1, 1, 1	
			Prediction horizon 24: 1, 1, 1	
			Prediction horizon 48: 1, 1, 1	
	Q	Seasonal moving average coefficient	Prediction horizon 4: 2, 3, 1	Obtained by Akaike information criterion (AIC)
			Prediction horizon 8: 3, 3, 2	
			Prediction horizon 24: 2, 3, 1	
			Prediction horizon 48: 3, 2, 3	
	D	Seasonal difference coefficient	Prediction horizon 4: 1, 1, 1	Obtained by Akaike information criterion (AIC)
			Prediction horizon 8: 1, 1, 1	
			Prediction horizon 24: 1, 1, 1	
			Prediction horizon 48: 1, 1, 1	
FA-LSSVM; LMD and FA-LSSVM	$t_{\max}$	The maximum number of iterations	100	Common value
	p	The population size	20	Common value
	ω	Randomness coefficient	0.5	Common value
	γ	Absorption coefficient	1	Common value
Improved FA-LSSVM; EEMD and improved FA-LSSVM	N_{imf}	The number of components	5	Obtained by EEMD
	$t_{\max}$	The maximum number of iterations	100	Common value
	p	The population size	20	Common value
	$\omega_{\max}$	Maximum randomness coefficient	1	Common value
	$\omega_{\min}$	Minimum randomness coefficient	0.01	Common value
	α_1	Convergence coefficient	0.8	Common value
	α_2	Convergence coefficient	5	Common value
	k	The ratio of white noise to the standard deviation of original signal amplitude	0.1	Common value

Table 4

The results of the correlation coefficient matrix.

Sequence	The correlation coefficient matrix
Corrcoef (PF1, original)	$\begin{bmatrix} 1 & -0.0375 \\ -0.0375 & 1 \end{bmatrix}$
Corrcoef (PF2, original)	$\begin{bmatrix} 1 & -0.1596 \\ -0.1596 & 1 \end{bmatrix}$
Corrcoef (residual, original)	$\begin{bmatrix} 1 & 0.3117 \\ 0.3117 & 1 \end{bmatrix}$

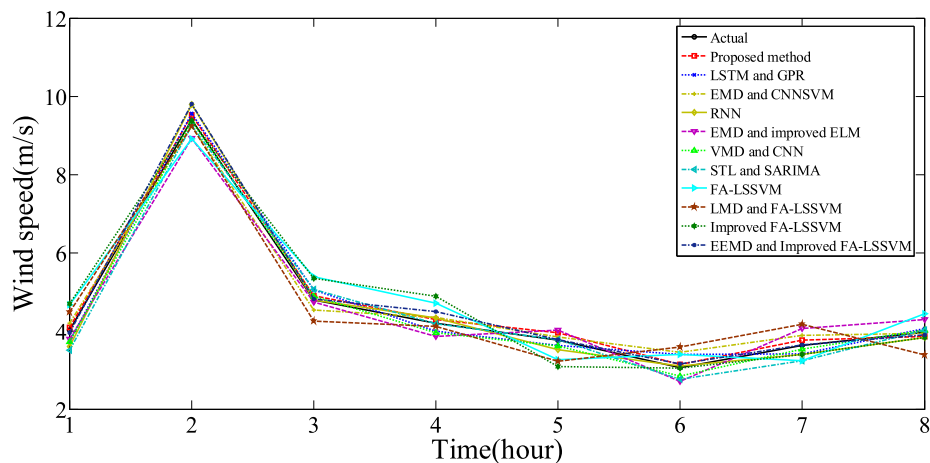


Fig. 11. The predictive comparison results of these prediction methods (prediction horizon is 8).

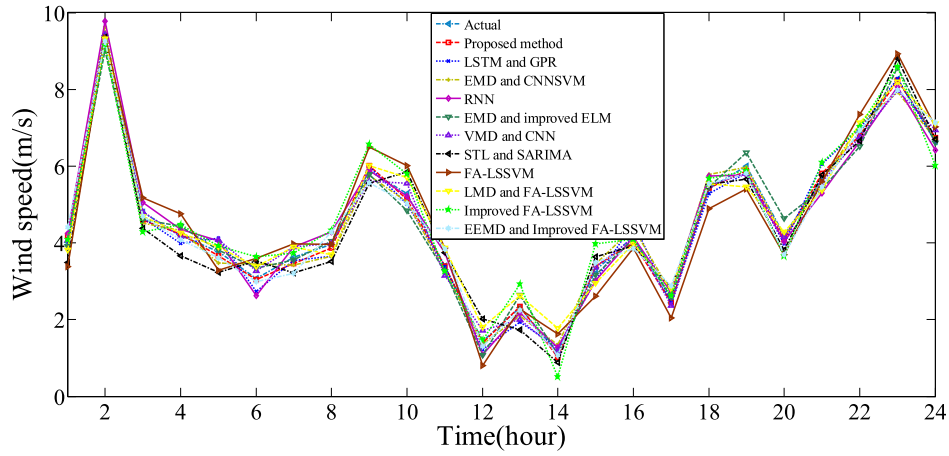


Fig. 12. The predictive comparison results of these prediction methods (prediction horizon is 24).

3.2. Online prediction process

Step 1 After the optimal parameters of each combined kernel function LSSVM model are obtained. The online multi-step prediction for short-term based on the proposed prediction model can be achieved. Assume the current sampling time is t , the short-term wind speed sequence is $y(t - N + 1), y(t - N + 2), \dots, y(t)$, the length of sequence is N .

Step 2 The short-term wind speed sequence is decomposed into k PF components with different scales and 1 residual component by LMD.

Step 3 For each PF component and residual component, the trained combined kernel function LSSVM model is used for prediction. Assume the PF component and residual component after decomposition is expressed as $x(t - N + 1), x(t - N + 2), \dots, x(t)$, the optimal m is used to determine the input of combined kernel function LSSVM. The input of combined kernel function LSSVM is express as Eq. (24).

$$X = \begin{bmatrix} x(t-m) & x(t-m-1) & \dots & x(t-1) & x(t) \end{bmatrix} \quad (24)$$

Step 4 X in Eq. (24) is chosen as the input of each combined kernel function LSSVM, then the predictive value Y can be obtained. Y is expressed as Eq. (25).

$$Y = \begin{bmatrix} \bar{x}(t+1) & \bar{x}(t+2) & \dots & \bar{x}(t+h-1) & \bar{x}(t+h) \end{bmatrix}^T \quad (25)$$

Step 5 Assume the predictive value of i th PF component is $\bar{x}_i(t+1), \bar{x}_i(t+2), \dots, \bar{x}_i(t+h)$, the predictive of residual component is $\bar{r}(t+1), \bar{r}(t+2), \dots, \bar{r}(t+h)$, then the final predictive value of short-term expressed as Eq. (26) is obtained. The final predictive values are $\bar{y}(t+1), \bar{y}(t+2), \dots, \bar{y}(t+h)$.

$$\begin{bmatrix} \bar{y}(t) \\ \bar{y}(t+1) \\ \vdots \\ \bar{y}(t+h) \end{bmatrix} = \begin{bmatrix} \bar{x}_1(t) \\ \bar{x}_1(t+1) \\ \vdots \\ \bar{x}_1(t+h) \end{bmatrix} + \begin{bmatrix} \bar{x}_2(t) \\ \bar{x}_2(t+1) \\ \vdots \\ \bar{x}_2(t+h) \end{bmatrix} + \dots + \begin{bmatrix} \bar{x}_k(t) \\ \bar{x}_k(t+1) \\ \vdots \\ \bar{x}_k(t+h) \end{bmatrix} + \begin{bmatrix} \bar{r}(t) \\ \bar{r}(t+1) \\ \vdots \\ \bar{r}(t+h) \end{bmatrix} \quad (26)$$

As shown in Eqs. (25) and (26), h can be any value within the allowable range of sample size. In the process of modeling, different h will generate different prediction models. This means that different prediction models are needed for different prediction horizon. The modeling process is the same, but the difference is that the parameters of different prediction models are generated after training.

The number of decomposition layers k of LMD is also an important parameter. The larger the value of k is, the higher the complexity of the model. However, unlike EMD or EEMD, the decomposition level of

LMD is determined by the characteristics of the signal itself. Through the iterative algorithm, different PF components are obtained until the termination condition is satisfied. Therefore, k does not need to be specified in the algorithm. Generally speaking, the decomposition level of LMD is less than EMD or EEMD, resulting in faster calculation time (Xie et al., 2016).

4. Experiments and analysis

4.1. Data description

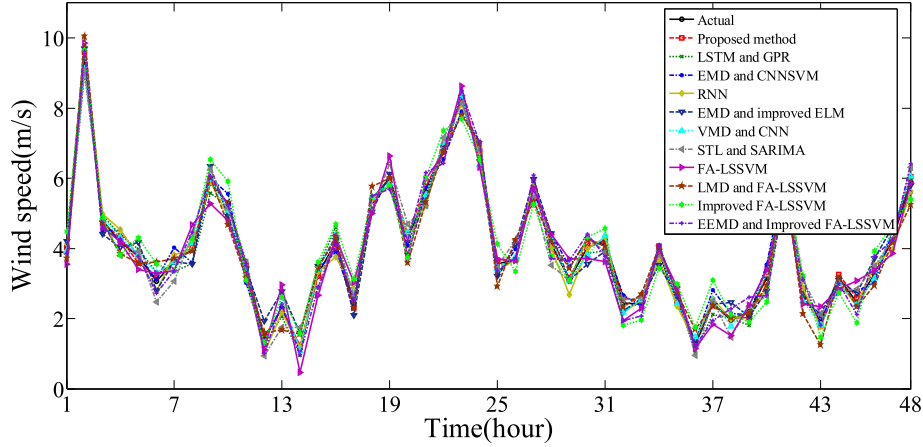
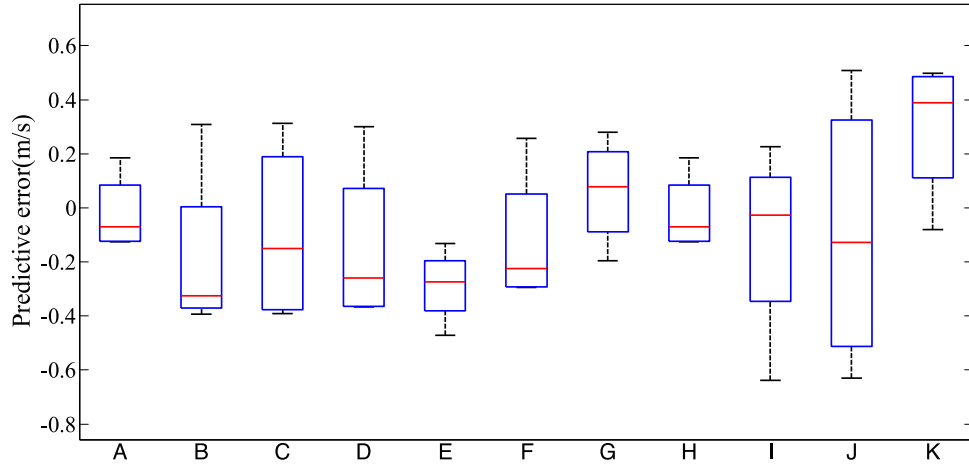
In this paper, five short-term wind speed datasets are evaluated to develop an effective wind speed prediction system for wind farms. The collected datasets are as shown in Fig. 3.

In order to demonstrate the advantages of small sample size in this paper, the length of each data set is 298. The total length of the sample is 1490. The 1st–298th sample is named as dataset A. The 299th–596th sample is named as dataset B. The 597th–984th sample is named as dataset C. The 895th–1192th sample is named as dataset D. The 1193th–1490th sample is named as dataset E. All these datasets with an hour sampling period and collected from a wind farm in Fotang Village, Fangshan Town, Heishan County, Jinzhou, Liaoning Province were collected. Location information is N41°53′54.50″, E122°03′44.41″. The altitude is 45 m. The collection site is located in the North Temperate Zone with Continental Climate, and the inland area is not along with seacoast. The collection area has four distinct seasons and abundant wind resources. The collection time is from 0 o'clock in Apr 1, 2016 to 1 o'clock in Jun 2, 2016. The former 250 sets of dataset A are used for model training. The following 48 sets of dataset A are used for model test. The dataset B to dataset E are used for to test the robustness of the proposed prediction method, and does not participate in the training process of the model. The descriptive statistical values of five datasets used in this study including the mean, standard deviation, maximum, minimum, Skewness, Kurtosis, median values, and sample size are listed in Table 2. The five datasets listed in Table 2 also present different characteristics. Meanwhile, 5-fold cross validation is used to improve the reliability of the prediction method. 250 short-term wind speed data is chosen as training set. 48 short-term wind speed data is chosen as test set. The 250 training set data is divided into 5 groups. At each time, 200 data are used for training and 50 data for testing. A total of 5 tests are conducted. RMSE is chosen as evaluating indicator. The minimum RMSE corresponding model is used as the final prediction model.

Table 5

The optimal parameters of each combined kernel function LSSVM with four prediction horizons.

Component	Prediction horizon	The optimal parameters
PF1	4	$a = 0.7215, q = 2, \sigma^2 = 25.3671, \gamma = 226.2560, m = 20$
	8	$a = 0.7458, q = 2, \sigma^2 = 127.3294, \gamma = 74.2381, m = 21$
	24	$a = 0.8682, q = 3, \sigma^2 = 329.3657, \gamma = 26.2571, m = 23$
	48	$a = 0.7962, q = 1, \sigma^2 = 28.3573, \gamma = 102.3647, m = 23$
PF2	4	$a = 0.7825, q = 1, \sigma^2 = 103.6971, \gamma = 42.3671, m = 18$
	8	$a = 0.7128, q = 1, \sigma^2 = 226.3201, \gamma = 3.6942, m = 22$
	24	$a = 0.7726, q = 2, \sigma^2 = 18.2697, \gamma = 237.3257, m = 22$
	48	$a = 0.7960, q = 2, \sigma^2 = 336.3210, \gamma = 26.3571, m = 24$
Residual	4	$a = 0.6027, q = 1, \sigma^2 = 77.3642, \gamma = 44.6840, m = 17$
	8	$a = 0.6113, q = 1, \sigma^2 = 108.3675, \gamma = 49.3641, m = 19$
	24	$a = 0.6445, q = 1, \sigma^2 = 31.3682, \gamma = 226.3641, m = 22$
	48	$a = 0.7030, q = 1, \sigma^2 = 10.3627, \gamma = 111.6428, m = 25$

**Fig. 13.** The predictive comparison results of these prediction methods (prediction horizon is 48).**Fig. 14.** The box-plot for the predictive error of each model (prediction horizon is 4; A: proposed method; B: LSTM and GPR; C: EMD and CNNSVM; D: RNN; E: EMD and improved ELM; F: VMD and CNN; G: STL and SARIMA; H: FA-LSSVM; I: LMD and FA-LSSVM; J: Improved FA-LSSVM; K: EEMD and Improved FA-LSSVM).

4.2. The performance indicators

In order to illustrate the effectiveness of the prediction method for short-term wind speed, the following 3 performance indicators are introduced to measure the prediction accuracy of the prediction methods (Li and Han, 2018).

1. Root mean square error (RMSE)

$$RMSE = \sqrt{\frac{1}{N} \sum_{t=1}^N (y(t) - \bar{y}(t))^2} \quad (27)$$

2. R^2 (R Square)

$$R^2 = 1 - \frac{\sum_{t=1}^N (y(t) - \bar{y}(t))^2}{\sum_{t=1}^N (y(t) - y_m)^2} \quad (28)$$

3. Reliability

$$R^{(1-a)} = \left[\frac{\xi^{(1-a)}}{N} - (1-a) \right] \times 100\% \quad (29)$$

where, N is number of sample, $y(t)$ is actual value of short-term wind speed time series, $\bar{y}(t)$ is prediction value of short-term wind speed, y_m is the mean value of short-term wind speed, $\xi^{(1-a)}$ is the number of actual values falling into the confidence interval under the confidence

Table 6

The comparison results of predictive error of these prediction models.

Prediction methods	Prediction horizon	Absolute maximum	Absolute minimum	Mean	Standard deviation	Variance
Proposed method	4	0.1857	0.1259	-0.0198	0.1456	0.0212
	8	0.0996	0.1839	-0.0875	0.0842	0.0071
	24	0.1952	0.1669	0.0371	0.1126	0.0127
	48	0.1878	0.1952	-0.0287	0.1032	0.0106
LSTM and GPR	4	0.3080	0.3935	-0.1837	0.3300	0.1089
	8	0.2828	0.3362	-0.0042	0.2378	0.0566
	24	0.3786	0.4126	0.0604	0.2217	0.0491
	48	0.4011	0.4292	0.0491	0.2183	0.0477
VMD and CNNSVM	4	0.3133	0.3905	-0.0940	0.3428	0.1175
	8	0.2630	0.3858	-0.0758	0.2151	0.0462
	24	0.3482	0.3880	-0.0036	0.2278	0.0519
	48	0.4036	0.4021	-0.0296	0.2546	0.0648
RNN	4	0.3013	0.3658	-0.1464	0.3141	0.0987
	8	0.4120	0.3962	0.0558	0.2534	0.0642
	24	0.4359	0.3819	-0.0115	0.2667	0.0712
	48	0.4390	0.4238	0.0489	0.2096	0.0439
EMD and improved ELM	4	0.1324	0.4706	-0.2882	0.1396	0.0195
	8	0.4656	0.4293	0.0538	0.3480	0.1211
	24	0.4503	0.4234	0.0200	0.2500	0.0625
	48	0.4642	0.4937	-0.0101	0.2959	0.0876
VMD and CNN	4	0.2576	0.2941	-0.1212	0.2607	0.0680
	8	0.2905	0.0936	0.1128	0.1411	0.0199
	24	0.3101	0.3175	-0.0101	0.1926	0.0371
	48	0.2915	0.3135	-0.0152	0.1706	0.0291
STL and SARIMA	4	0.2791	0.1962	0.0593	0.2009	0.0404
	8	0.5076	0.2638	0.1266	0.2577	0.0664
	24	0.5758	0.5747	0.0728	0.4030	0.1624
	48	0.6215	0.6086	0.0116	0.3540	0.1253
FA-LSSVM	4	0.1857	0.1259	-0.0198	0.1456	0.0212
	8	0.0996	0.1839	-0.0875	0.0842	0.0171
	24	0.1952	0.1669	0.0371	0.1126	0.0137
	48	0.1878	0.1952	-0.0287	0.1032	0.0136
LMD and FA-LSSVM	4	0.2259	0.6393	-0.1167	0.3689	0.1361
	8	0.5789	0.5505	0.0456	0.4975	0.2475
	24	0.5400	0.6255	-0.0952	0.3019	0.0912
	48	0.6537	0.6584	0.0585	0.3544	0.1256
Improved FA-LSSVM	4	0.5092	0.6295	-0.0939	0.5150	0.2652
	8	0.6818	0.6936	-0.1051	0.4898	0.2399
	24	0.6390	0.6952	-0.1019	0.4079	0.1664
	48	0.6648	0.6846	-0.0692	0.4381	0.1920
EEMD and Improved FA-LSSVM	4	0.4973	0.0802	0.2984	0.2666	0.0711
	8	0.0726	0.4075	-0.0884	0.1742	0.0304
	24	0.5228	0.4879	0.0388	0.2859	0.0817
	48	0.4997	0.5294	0.0082	0.3195	0.1021

level $1 - \alpha$. Given confidence level $1 - \alpha$, α is from 1% to 100%. If the relative error of the prediction method is smaller than or equal to $1 - \alpha$, then reliability plus 1. Otherwise, reliability stays the same. The purpose of introducing reliability is to evaluate the reliability of confidence interval. Furthermore, the Pearson's test is used to test prediction accuracy from the statistical perspective. Pearson's test can measure the association strength between the actual value and the predictive value.

Finally, in order to prove the prediction performance, the Diebold-Mariano (DM) test is used to test prediction accuracy from the hypothesis test. The definition of DM test is as the follows. The hypothesis tests are:

$$H_0 : E(d_h) = 0, \forall n \quad (30)$$

$$H_1 : E(d_h) \neq 0, \exists n \quad (31)$$

The DM test statistic values equals the next.

$$DM = \frac{\sum_{h=1}^k (L(\varepsilon_{t+h}^{(A)}) - L(\varepsilon_{t+h}^{(B)})) / k}{\sqrt{S^2 / k}} \quad (32)$$

where ε_{t+h} is the prediction error, S^2 is an estimation value for the variance of $d_h = L(\varepsilon_{t+h}^{(A)}) - L(\varepsilon_{t+h}^{(B)})$, and L is the loss function, which is

performed to measure the prediction accuracy. Two common versions of L are widely used in the studies include absolute deviation error loss and square error loss. The test statistic DM is convergent to the standard normal distribution. The null hypothesis can be rejected if the following formula is satisfied.

$$|DM| > z_{\alpha/2} \quad (33)$$

where $z_{\alpha/2}$ is the critical z -value and α is the significance level. Under the assumption that the two models have the same predictive performance, the loss functions of the two models have equal unconditional expectations, that is

$$E(d_h) = E(L(\varepsilon_1) - L(\varepsilon_2)) \quad (34)$$

Some analysis and discussion are made on these performance indicators.

RMSE. The smaller the RMSE is, the closer the sample value is to the average value, that is to say, the less discrete the predicted values of the prediction model is.

R^2 . The closer it approaches 1, the stronger the interpretation ability of the prediction model to the actual data, that is, the better the model fits the data.

The Pearson's test. If Pearson's correlation coefficient is equal to 1, it indicates that the actual value and the predictive value have a linear

Table 7

Comparison of performance indicators of prediction methods for short-term wind speed.

Prediction methods	Prediction horizon	RMSE	R^2
Proposed method	4	0.1060	0.9967
	8	0.1163	0.9965
	24	0.1177	0.9964
	48	0.1276	0.9961
LSTM and GPR	4	0.2216	0.9875
	8	0.2225	0.9868
	24	0.2253	0.9784
	48	0.3397	0.9838
VMD and CNNSVM	4	0.2150	0.9870
	8	0.2230	0.9786
	24	0.2537	0.9850
	48	0.3114	0.9836
RNN	4	0.2131	0.9858
	8	0.2435	0.9855
	24	0.2614	0.9824
	48	0.3089	0.9752
EMD and improved ELM	4	0.2455	0.9839
	8	0.2930	0.9819
	24	0.3126	0.9711
	48	0.3299	0.9641
VMD and CNN	4	0.1695	0.9908
	8	0.1737	0.9902
	24	0.1888	0.9899
	48	0.2562	0.9880
STL and SARIMA	4	0.1838	0.9834
	8	0.2723	0.9797
	24	0.3505	0.9611
	48	0.4011	0.9603
FA-LSSVM	4	0.3940	0.9604
	8	0.5025	0.9163
	24	0.4867	0.9476
	48	0.4027	0.9507
LMD and FA-LSSVM	4	0.3401	0.9817
	8	0.4676	0.9339
	24	0.3105	0.9725
	48	0.3555	0.9595
Improved FA-LSSVM	4	0.4558	0.9416
	8	0.4700	0.9410
	24	0.4121	0.9566
	48	0.4390	0.9390
EEMD and Improved FA-LSSVM	4	0.3773	0.9690
	8	0.1854	0.9913
	24	0.2825	0.9787
	48	0.3162	0.9702

relationship. On the other hand, if Pearson's correlation coefficient is equal to 0, there is no relationship between actual value and prediction value.

The DM test. If $E(d_h) > 0$, it means the predictive performance of model 2 is better than model 1. If $E(d_h) = 0$, it means the predictive performance of model 1 is the same as that of model 2. Otherwise, the predictive performance of model 1 is better than model 2.

4.3. Experimental setup

In order to verify the prediction effect, some short-term wind speed prediction methods are chosen to compare. Six prediction models include LSTM and GPR (Zhang et al., 2019a), EMD and CNNSVM (Mi et al., 2019), RNN (Li et al., 2019), EMD and improved ELM (Tian et al., 2018c), VMD and CNN (Zhou et al., 2018), and seasonal-trend decomposition based on loess (STL) and seasonal autoregressive integrated moving average (SARIMA). In addition, the proposed prediction method is also compared with the improved FA-LSSVM, LMD and FA-LSSVM, FA-LSSVM, and EEMD and improved FA-LSSVM, which to identify that using the LMD decomposition as well as the improved FA improving the accuracy of the results. The description of these parameters is given in Table 3. Meanwhile, how to determine these parameters are also concluded (Ribeiro and Coelho, 2020).

The following explains how to determine the parameters of the prediction methods in Table 3.

LSTM and GPR: The idea for combination is to first train LSTM network completely, then use the training set and test set as inputs to make the first prediction with LSTM, finally construct GPR for the second prediction between the first predicted value and the observed value.

CNNSVM: In the CNNSVM, the hyper-parameters of the convolutional layers are fixed, while the parameters of the SVM layer can be optimized through the grid search algorithm.

RNN: the units of LSTM layer slightly exceed the maximum cell stages to cover all continuous time points in updating parameters of LSTM. Then the first fully connected layer is used to pre-train each pair of features under supervised situation, then the drop out layer is applied to reduce over fitting on pre-train stage.

Improved ELM: It can randomly select its training parameters, and get a complete network training model only through the output weights obtained by the least squares method. The new sample should be added to the training set after the initial network weights are calculated, and the corresponding network weights can be obtained on the basis of the initial network weights. At the same time, the new and old samples are given different weights.

Table 8

Results of the Pearson's test of these prediction methods for short-term wind speed.

Prediction methods	Prediction horizon	Pearson's correlation coefficient
Proposed method	4	0.9992
	8	0.9984
	24	0.9984
	48	0.9983
LSTM and GPR	4	0.9945
	8	0.9938
	24	0.9930
	48	0.9923
VMD and CNNSVM	4	0.9959
	8	0.9949
	24	0.9935
	48	0.9895
RNN	4	0.9987
	8	0.9967
	24	0.9932
	48	0.9912
EMD and improved ELM	4	0.9985
	8	0.9922
	24	0.9862
	48	0.9857
VMD and CNN	4	0.9979
	8	0.9966
	24	0.9954
	48	0.9953
STL and SARIMA	4	0.9972
	8	0.9921
	24	0.9806
	48	0.9804
FA-LSSVM	4	0.9890
	8	0.9672
	24	0.9751
	48	0.9755
LMD and FA-LSSVM	4	0.9977
	8	0.9685
	24	0.9891
	48	0.9801
Improved FA-LSSVM	4	0.9887
	8	0.9716
	24	0.9795
	48	0.9698
EEMD and Improved FA-LSSVM	4	0.9948
	8	0.9984
	24	0.9897
	48	0.9856

CNN: The four channels in input data match the four decomposed modes from wind speed signal. Then, multiple filters extract features in different scale to capture the mapping relationship between input data and output data. The convolution of stride is employed instead of pooling because stride convolution is fully differentiable.

STL and SARIMA: The period length of the sequence is obtained by periodic analysis. The stability of the time series is identified, that is, the values of d and D are determined. Determine the approximate values of the remaining p , q , P and Q of the model, and adjust the values through AIC criteria. The maximum likelihood estimation method is used to determine the estimated values of all parameters.

FA-LSSVM: The LSSVM is chosen as the prediction model. And, FA is adopted as the optimization algorithm. The three parameters include σ^2 , γ , and m of LSSVM are determined by FA.

LMD and FA-LSSVM: The modeling of the prediction model is as the same with FA-LSSVM. The difference is that the original short-term wind speed is decomposed into various components through LMD. These components are predicted by FA-LSSVM.

Improved FA-LSSVM: The LSSVM is chosen as the prediction model. And, improved FA is adopted as the optimization algorithm. The three parameters include σ^2 , γ , and m of LSSVM are determined by improved FA.

EEMD and Improved FA-LSSVM: The modeling of the prediction model is as the same with improved FA-LSSVM. The difference is that the original short-term wind speed is decomposed into various components through EEMD. These components are predicted by improved FA-LSSVM. In EEMD algorithm, the white noise is added. The ratio of white noise to the standard deviation of original signal amplitude is set as 0.1. The standard deviation of short-term wind speed is 2.6331. Therefore the white noise amplitude is 0.2633.

4.4. Experimental comparison

250 sets short-term wind speed in dataset A is processed by LMD method, then 2 PF components and 1 residual component are obtained. The original short-term wind speed, 2 PF components, and 1 residual component are as shown in Fig. 4.

Fig. 4 shows that each PF component contains the different characteristic information in the original short-term wind speed time series. Meanwhile, the correlation coefficient matrix between PF1, PF2, residual and the original short-term wind speed series are calculated. The results of the correlation coefficient matrix are shown in Table 4.

It can be seen from Fig. 4 and Table 4, PF1 component changes dramatically, and the correlation coefficient between it and original

Table 9

The DM test statistical index table of proposed method with each compared model (prediction horizon is 4).

Model	1% significance level	5% significance level	10% significance level
LSTM and GPR	4.6957	1.1461	1.4609
VMD and CNNSVM	2.3911	1.1224	1.6513
RNN	2.6086	1.1274	1.3491
EMD and improved ELM	1.6727	1.0915	1.1927
VMD and CNN	2.6086	1.1274	1.3491
STL and SARIMA	1.6727	1.0915	1.1927
FA-LSSVM	5.3669	2.3691	1.6195
LMD and FA-LSSVM	5.3760	2.4336	1.7659
Improved FA-LSSVM	9.6163	2.4864	1.7193
EEMD and Improved FA-LSSVM	6.6321	2.3381	1.5955

wind speed is weak, which is expressed as the main fluctuation component of the original short-term wind speed. The sequence variation of PF2 component is weak and has some micro correlation between it and original wind speed. There is a strong correlation between the residual component and the original data, which contains the important deformation information of the original short-term wind speed series. Each component after decomposition contains basically the same deformation information, which reduces the interference between different information in modeling, and the change curve of each component decomposed is smoother than the original sequence curve. Therefore, the prediction model for each component will reduce the difficulty of modeling and improve the prediction accuracy.

250 sets PF components and residual component are used to train 3 combined kernel function LSSVM models. Improved FA is introduced to optimize the parameters of 3 combined kernel function LSSVM models. The parameters of improved FA are as the follows: the maximum number of iterations $t_{\max}$ is 100, the population size is 20, $\omega_{\max}$ is 1, $\omega_{\min}$ is 0.01, α_1 is 0.8, α_2 is 5, $\alpha_{\max}$ is 1, and $\alpha_{\min}$ is 0.01. The range of parameters are as $a \in [0, 1]$, $q \in [1, 5]$, $\sigma^2 \in [0.01, 1000]$, $\gamma \in [0.01, 1000]$, and $m \in [1, 30]$. In many similar studies, the prediction horizon (prediction step) is generally chosen as 1, 2, 4, 8, 12, 16, 24, 30, 36, and 48. Therefore, the prediction horizon h is chosen as 4, 8, 24 and 48. Because different prediction horizons only affect the parameters of the established prediction model, these four prediction steps can show the effectiveness of the prediction model. Before optimization, data need to be normalized. At the same time, 5-fold cross validation method is used to improve the generalization ability of the model. After training, 5 prediction methods are obtained. In each training process, improved FA is used to optimize the parameters. The parameters with the minimum fitness value are selected as the final optimization parameters. The fitness curves of improved FA optimized three components in four prediction horizons with minimum fitness are as shown in Fig. 5. After optimization, the optimal of parameters of each combined kernel function LSSVM is as shown in Table 5. As shown in Table 5, for the same prediction horizon, three different models are generated for each PF component and residual component.

The predictive value of each combined kernel function LSSVM model can be obtained. The comparison curves between the actual values and the predictive values of each component under four prediction horizons are shown in Figs. 6–9. From these figures, we can know that the obtained combined kernel function LSSVM model has good prediction effect on PF1, PF2 and residual component.

When the predictive values of PF components and residual component are obtained, the final predictive results can be obtained by superimposing these predictive values. The final predictive values are shown in Figs. 10–13. In order to compare the prediction accuracy, the ten prediction methods mentioned in Section 4.3 are selected as the comparison models. The predictive results of these prediction methods are also shown in Figs. 10–13.

It can be observed from Figs. 10–13 that the proposed prediction method is superior to the other prediction methods. The predicted value of this method is closer to the actual short-term wind speed data. It can be seen that the proposed prediction method has better prediction

ability for short-term wind speed. Figs. 14–17 give the box-plot for the predictive error of each model used for predictions. The results in these figures show that the proposed prediction method reduces the predictive error in relation to other comparative models. It can be known that the proposed prediction method with smaller predictive error and smaller variation. The proposed prediction method shows to be stable.

Table 6 gives the comparison results include absolute maximum, absolute minimum, mean, standard deviation, and variance of relative predictive error of these prediction models. It can be seen from Table 6, the absolute maximum, absolute minimum, standard deviation, and variance of relative predictive error of the proposed prediction method is smaller than that of the other comparative models. Namely, the relative error can better reflect the credibility of the proposed prediction method.

Table 7 gives RMSE and R^2 comparison of these prediction methods for short-term wind speed. From the performance indicators comparison results in Table 7, RMSE of the proposed prediction method is smaller than the other prediction methods. At the same time, R^2 value of the proposed prediction method is closer to 1 than the other prediction methods. The closer the value of R^2 is to 1, the better the regression prediction performance of the model is. Therefore, the prediction accuracy of the proposed prediction method for short-term wind speed is better than the other prediction methods.

Figs. 18–21 are the reliability and confidence distribution with different prediction horizons of the prediction methods mentioned in this paper. It can be observed from this graph that the prediction method in this paper has higher reliability under the same confidence level. The prediction value of short-term wind speed can effectively track the variation of wind speed series with high confidence. The prediction value of short-term wind speed of the proposed prediction method falls more in the same confidence interval, which shows the validity of the probability prediction results.

The results of Pearson's test are given in Table 8. From Table 8, it can be observed that the results of Pearson's test of the proposed prediction method are higher than those of the other prediction methods. The results show that the association strength between the actual value and the prediction value of the short-term wind speed of the proposed prediction method is stronger than the other prediction methods.

The DM test values calculated by the square error loss function with four prediction horizons are listed in Tables 9–12. From the result in Tables 9–12, the DM test values between other models and the proposed model are greater than 0, the proposed prediction model significantly outperforms the other prediction models at the same significance level. Thus, it can reasonably be concluded that the proposed prediction method is superior to the other models.

In order to specifically describe the prediction results of different prediction methods, a Taylor diagram is employed and illustrated in Figs. 22–25. These 4 figures are corresponding to the prediction horizon of 4, 8, 24 and 48, respectively. The standard deviations, root mean square deviation (RMSD) and the correlation coefficient of predictive value between each prediction method and observation (actual) value of short-term wind speed are clearly exhibited. In these figures, the

Table 10

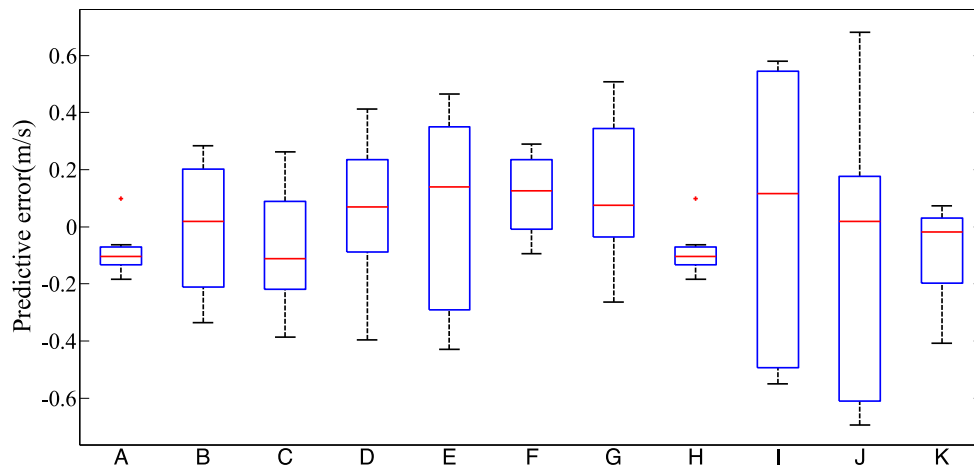
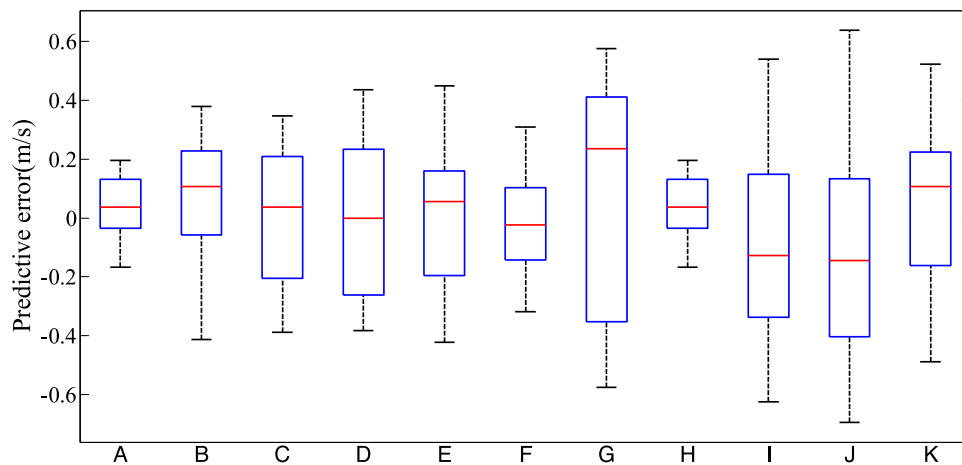
The DM test statistical index table of proposed method with each compared model (prediction horizon is 8).

Model	1% significance level	5% significance level	10% significance level
LSTM and GPR	2.4787	1.2506	1.0568
VMD and CNNSVM	1.7249	1.1314	1.0443
RNN	1.8812	1.1625	1.0481
EMD and improved ELM	3.8338	1.2493	1.0639
VMD and CNN	1.8812	1.1625	1.0481
STL and SARIMA	3.8338	1.2493	1.0639
FA-LSSVM	3.3818	1.6690	2.4530
LMD and FA-LSSVM	3.9401	1.5107	2.5815
Improved FA-LSSVM	4.5056	1.7663	2.7335
EEMD and Improved FA-LSSVM	3.8964	1.7107	2.7652

Table 11

The DM test statistical index table of proposed method with each compared model (prediction horizon is 24).

Model	1% significance level	5% significance level	10% significance level
LSTM and GPR	3.3818	1.6690	1.1609
VMD and CNNSVM	3.7722	1.8211	1.3192
RNN	5.0418	1.8117	1.3134
EMD and improved ELM	3.5402	1.6796	1.2926
VMD and CNN	5.0418	1.8117	1.3134
STL and SARIMA	3.5402	1.6796	1.2926
FA-LSSVM	4.6957	1.1461	1.4609
LMD and FA-LSSVM	0.9626	0.9902	1.0344
Improved FA-LSSVM	2.2691	1.1190	1.2424
EEMD and Improved FA-LSSVM	2.2599	1.1188	1.8181

**Fig. 15.** The box-plot for the predictive error of each model (prediction horizon is 8; A: proposed method; B: LSTM and GPR; C: EMD and CNNSVM; D: RNN; E: EMD and improved ELM; F: VMD and CNN; G: STL and SARIMA; H: FA-LSSVM; I: LMD and FA-LSSVM; J: Improved FA-LSSVM; K: EEMD and Improved FA-LSSVM).**Fig. 16.** The box-plot for the predictive error of each model (prediction horizon is 24; A: proposed method; B: LSTM and GPR; C: EMD and CNNSVM; D: RNN; E: EMD and improved ELM; F: VMD and CNN; G: STL and SARIMA; H: FA-LSSVM; I: LMD and FA-LSSVM; J: Improved FA-LSSVM; K: EEMD and Improved FA-LSSVM).

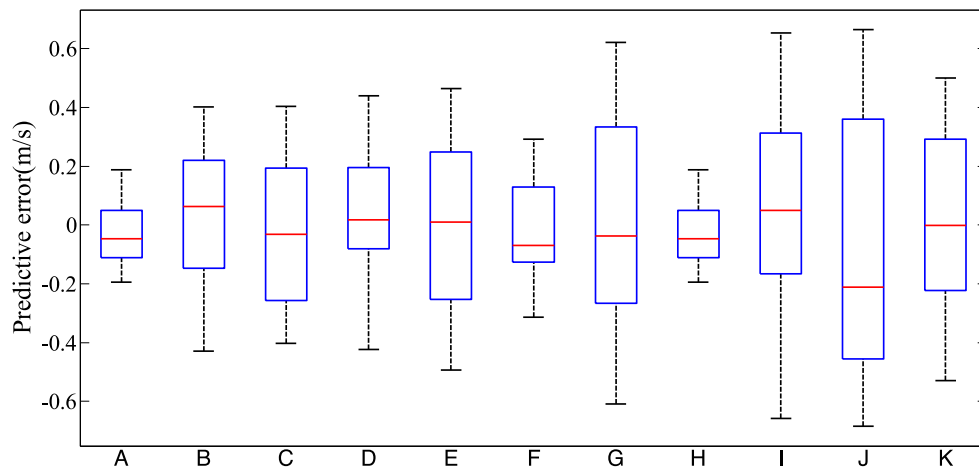


Fig. 17. The box-plot for the predictive error of each model (prediction horizon is 48; A: proposed method; B: LSTM and GPR; C: EMD and CNNSVM; D: RNN; E: EMD and improved ELM; F: VMD and CNN; G: STL and SARIMA; H: FA-LSSVM; I: LMD and FA-LSSVM; J: Improved FA-LSSVM; K: EEMD and Improved FA-LSSVM).

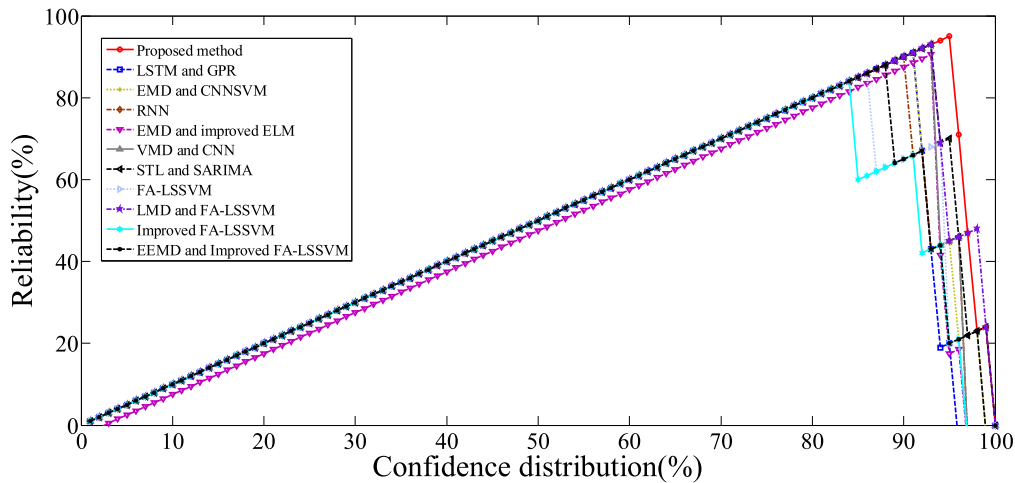


Fig. 18. Reliability and confidence distribution of these prediction methods (prediction horizon is 4).

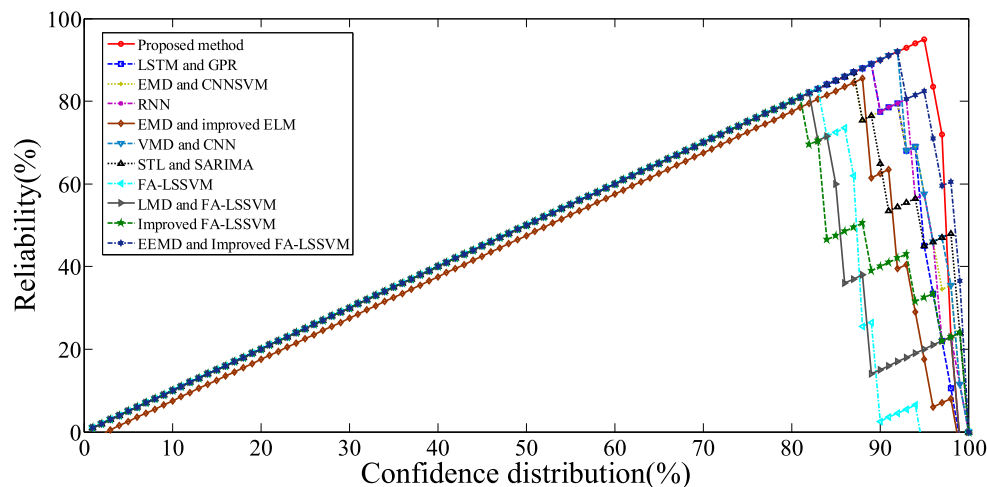


Fig. 19. Reliability and confidence distribution of these prediction methods (prediction horizon is 8).

horizontal and vertical coordinates represent standard deviation. The sector curve represents the correlation coefficient. The virtual curve represents RMSD. The dot A means observation (actual) value of short-term wind speed. The other dots represent the standard deviations, RMSD and correlation coefficient of the proposed prediction method

and other comparative models. It can be seen from these figures, dot B is more close to dot A. Therefore, the correlation coefficient of the prediction method is larger than other comparative models, which means that the predicted value of the proposed prediction method and the observation (actual) value fit better. In addition, the results in these

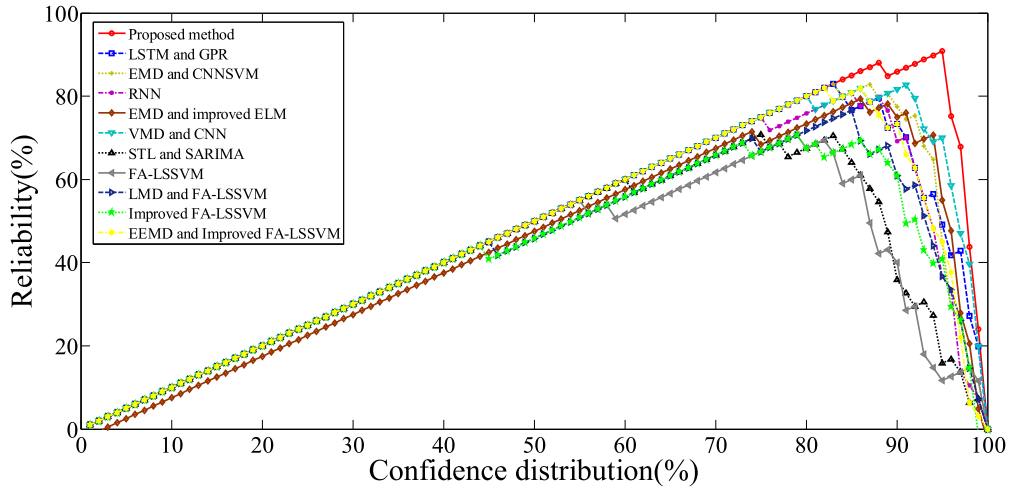


Fig. 20. Reliability and confidence distribution of these prediction methods (prediction horizon is 24).

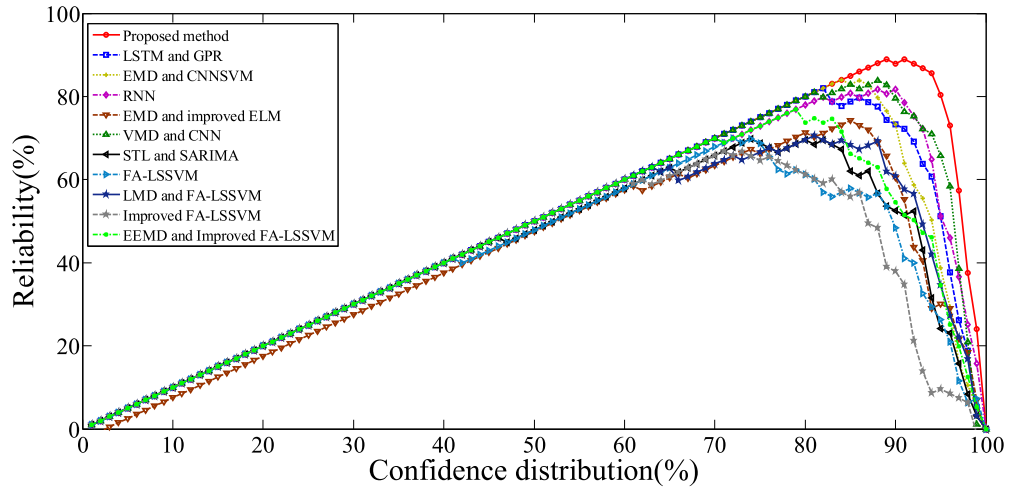


Fig. 21. Reliability and confidence distribution of these prediction methods (prediction horizon is 48).

Table 12

The DM test statistical index table of proposed method with each compared model (prediction horizon is 48).

Model	1% significance level	5% significance level	10% significance level
LSTM and GPR	5.3760	2.3691	1.6195
VMD and CNNSVM	6.6328	2.2544	1.7378
RNN	4.0946	2.2982	1.6368
EMD and improved ELM	6.4986	2.3555	1.6660
VMD and CNN	4.0946	2.2982	1.6368
STL and SARIMA	6.4986	2.3555	1.6660
FA-LSSVM	2.4787	1.2506	1.9878
LMD and FA-LSSVM	4.4693	1.2173	2.0771
Improved FA-LSSVM	2.6163	1.0649	1.6162
EEMD and Improved FA-LSSVM	0.9374	0.8845	1.1420

figures also show that proposed method has the smallest RMSD and the same standard deviation as the observation (actual) value. In general, the proposed prediction method has better performance.

4.5. Experiment of robustness

In order to test the robustness of the proposed method, two kinds of simulation experiments have been added. The first kind of experiment is to verify whether the model has stable prediction effect by using dataset B, C, D, and E. The second kind of experiment is to test the prediction ability by adding some interference noise to the input data of the model.

4.5.1. The first experiment

The parameters of the proposed prediction model are trained by dataset A. Four new test data sets are used to test the robustness of the prediction model while the prediction model remains unchanged. RMSE was selected as the evaluation index. Table 13 gives the experimental results. As can be seen from Table 13, the difference between RMSE of dataset A with other four new test sets is not very big, which shows that the proposed prediction model has strong robustness to the new wind speed data.

4.5.2. The second experiment

For data set A, the random noise signal whose amplitude is 5% of the original wind speed value is added. The dataset A after adding noise

Table 13
RMSE value of Dataset A and other four new data sets.

Dataset	Prediction horizon	RMSE
Dataset A	4	0.1060
	8	0.1163
	24	0.1177
	48	0.1276
Dataset B	4	0.1067
	8	0.1156
	24	0.1203
	48	0.1240
Dataset C	4	0.0973
	8	0.1164
	24	0.1223
	48	0.1259
Dataset D	4	0.0920
	8	0.1074
	24	0.1145
	48	0.1247
Dataset E	4	0.1085
	8	0.1092
	24	0.1189
	48	0.1248

Table 14
The RMSE results of dataset A and dataset A after adding noise.

Dataset	Prediction horizon	RMSE
Dataset A	4	0.1060
	8	0.1163
	24	0.1177
	48	0.1276
Dataset A after adding noise	4	0.1086
	8	0.1098
	24	0.1169
	48	0.1204

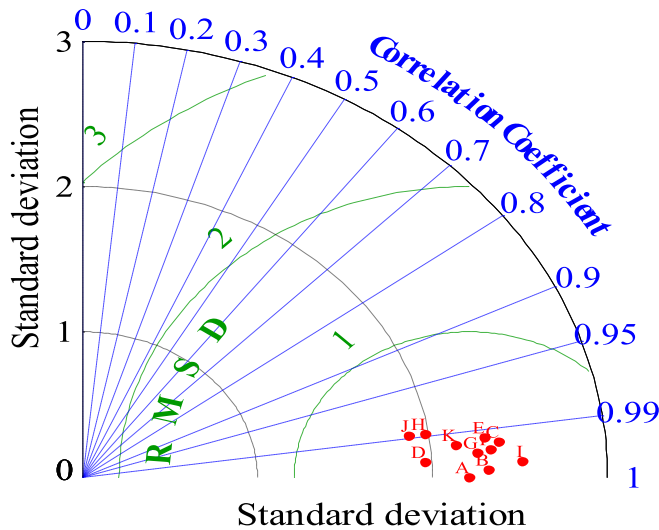


Fig. 22. Taylor diagram of the prediction methods with prediction horizon is 4 (A: actual; B: proposed method; C: LSTM and GPR; D: EMD and CNN; E: RNN; F: EMD and improved ELM; G: VMD and CNN; H: STL and SARIMA; I: FA-LSSVM; J: LMD and FA-LSSVM; K: Improved FA-LSSVM; L: EEMD and Improved FA-LSSVM).

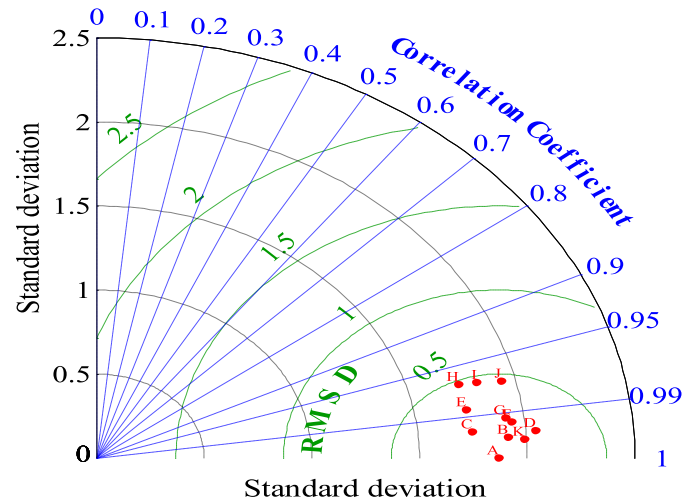


Fig. 23. Taylor diagram of the prediction methods with prediction horizon is 8 (A: actual; B: proposed method; C: LSTM and GPR; D: EMD and CNN; E: RNN; F: EMD and improved ELM; G: VMD and CNN; H: STL and SARIMA; I: FA-LSSVM; J: LMD and FA-LSSVM; K: Improved FA-LSSVM; L: EEMD and Improved FA-LSSVM).

signal is used to verify the robustness of the proposed prediction model to noise. RMSE is also chosen as the evaluation index. Table 14 gives the RMSE results of dataset A and dataset A after adding noise. As can be seen from Table 14, the difference between RMSE of dataset A and dataset A after adding noise is small, which shows that the proposed prediction model has strong robustness to the noise signal.

5. Discussion

In this section, the performance indicators of the developed prediction method are discussed. Then, the association strength based on Pearson's test is discussed, and the statistical significance is also evaluated by using the DM test. Next, the robustness of the proposed prediction method is discussed. Finally, the main advantages of the proposed method are given.

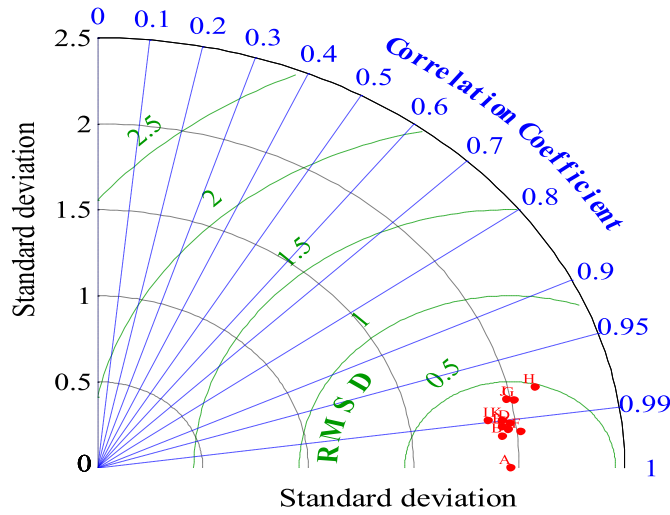


Fig. 24. Taylor diagram of the prediction methods with prediction horizon is 24 (A: actual; B: proposed method; C: LSTM and GPR; D: EMD and CNNSVM; E: RNN; F: EMD and improved ELM; G: VMD and CNN; H: STL and SARIMA; I: FA-LSSVM; J: LMD and FA-LSSVM; K: Improved FA-LSSVM; L: EEMD and Improved FA-LSSVM).

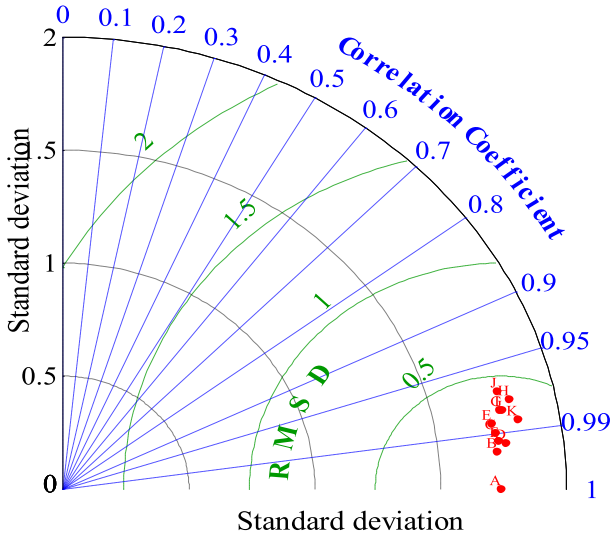


Fig. 25. Taylor diagram of the prediction methods with prediction horizon is 48 (A: actual; B: proposed method; C: LSTM and GPR; D: EMD and CNNSVM; E: RNN; F: EMD and improved ELM; G: VMD and CNN; H: STL and SARIMA; I: FA-LSSVM; J: LMD and FA-LSSVM; K: Improved FA-LSSVM; L: EEMD and Improved FA-LSSVM).

5.1. Discussion I: the performance indicators

To prove the superiority of the proposed prediction method thoroughly, the performance indicators between the proposed method and the compared models are discussed. Table 7 shows that: (a) RMSE value of the proposed prediction method is 5% to 25% of that of other methods. (b) R^2 value of the proposed prediction method is closer to 1 than the other prediction methods. The proposed prediction method has better performance indicators. The box-plot results of Figs. 14–17 show that the developed method has an absolute prediction error with a smaller variation, showing to be stable. Meanwhile, it can be concluded from Table 6 that the proposed method has smaller relative prediction error. Furthermore, the results in Figs. 18–21 shows that the proposed prediction method is more reliable under the same confidence level. All performance indicators not only reveals that the developed method is superior to the compared models in wind speed prediction, but also clearly explains the reason why the developed method shows

a given prediction performance. The reason for the improvement of all performance indicators is that the prediction method proposed can reveal the internal variation law of short-term wind speed better.

5.2. Discussion II: the association strength

In this section, the discussion of association strength based on Pearson's test is conducted to further prove the superiority of the developed prediction method compared with other models. In detail, if Pearson's correlation coefficient is equal to 1, it indicates that the actual data and the predictive data have a linear relationship, whereas, if Pearson's correlation coefficient is equal to 0, there is no relationship between these two datasets. The results of Pearson's test are presented in Table 8. From Table 8, it can be observed that the results of Pearson's test of the developed prediction method are higher than those of the other benchmark models in all cases.

5.3. Discussion III: the statistical significance

The statistical significance of the proposed prediction method is discussed to prove superiority in short-term wind speed prediction from the point of statistics. The DM test based on the square error loss function is used to show the significance of the superiority of the proposed prediction method compared with other models. The results for the DM test are listed in Tables 9–12, which indicates that the DM test values of all compared models are greater than the critical value of the 1%, 5%, and 10% significance level. Therefore, we can reject the null hypothesis at the 1%, 5%, and 10% significance level, and we believe that the proposed prediction method significantly outperforms all compared models. Therefore, we can reasonably draw the conclusion that the proposed prediction method not only achieves higher prediction accuracy, but also shows a significant difference in terms of the prediction accuracy level, which further proves the superiority of the developed proposed prediction method in short-term wind speed prediction.

5.4. Discussion IV: robustness

Two robustness experiments include new four datasets and dataset A of randomness noise signal with amplitude of 5% is added are performed out. In two cases, the parameters of the proposed prediction method stay the same. RMSE is chosen as the evaluation index to measure the robustness of the proposed method. The results in Tables 13 and 14 show that the RMSE of the two cases has little change, indicating that the established model can adapt to the new and noisy data set. Therefore, the proposed prediction method has strong robustness performance.

5.5. Discussion V: the advantages of the proposed method

From the above discussion, we can know that the proposed prediction method has achieved better results. According to Discussions I, II, III, and IV, we can safely draw the conclusion that the proposed prediction method shows a more powerful forecasting ability than the compared models. The main advantages of the proposed method in relation to compared models are as follows. (a) The introduction of LMD. After LMD processing, each component after decomposition contains basically the same deformation information, which reduces the interference between different information in modeling process. The prediction difficulty and complexity are both reduced. (b) The proposed combined kernel function LSSVM model has low dependence on single kernel functions and their parameters, and has high modeling accuracy. Combined with developed improved FA, the combined kernel function LSSVM can better describe the change law of the prediction object. (c) Because the developed prediction model is derived from the improvement of LSSVM, it inherits the advantages of LSSVM. Namely, the modeling process needs little sample data. This is very useful for the research object with limited sample size in practical application.

6. Conclusion

In this paper, LMD algorithm is used to decompose the original short-term wind speed time series. The PF components and residual component of different scales can be obtained, which simplifies the coupling between different feature information and reduces the difficulty of prediction and modeling. The combined kernel function is selected as the kernel function of LSSVM. The global kernel function is used to fit the correlation of the distant sample data. The generalization performance of the LSSVM model is improved. An improved FA with better convergence performance is introduced to optimize the parameters of combined kernel function LSSVM. The proposed prediction method is applied to the multi-step prediction of short-term wind speed time series collected from a wind farm, and compared with other prediction methods. The main results of this study are as the follows. (a) Compared with other state-of-the-art models, the proposed prediction method has better performance indicators. It shows that the proposed prediction method can better fit the short-term wind speed data. (b) From the statistical perspective, the DM test shows that the proposed prediction method significantly outperforms the other comparative models at different significance levels. (c) The Pearson's test of the proposed prediction method is higher than those of other comparative models. (d) The result of Taylor diagram indicates that the proposed method has smaller correlation coefficient as the observation. (e) Through two case studies, the robustness of the proposed prediction method is also verified. In a conclusion, the proposed prediction method is effective.

The short-term wind speed prediction method proposed in this paper has good application value in the engineering practice of wind system. However, the traditional LMD employs the moving average method to obtain the local mean function and local envelope function, which cause the over-smoothness easily and affects the precision of decomposition. Therefore, in the future, this paper will further study the improvement of LMD method to achieve accurate decomposition of short-term wind speed time series.

CRedit authorship contribution statement

Zhongda Tian: Conceptualization, Methodology, Software, Validation, Writing - original draft, Writing - review & editing.

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